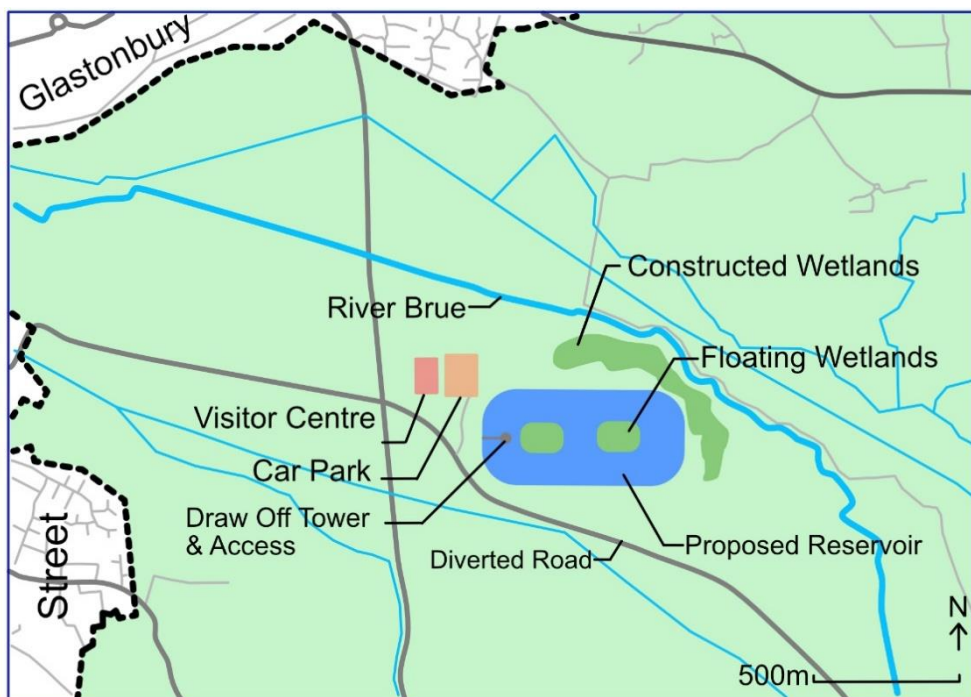




**engineering
4confluence**

Group 7

Glastonbury Reservoir Proposal



AI Acknowledgements

We acknowledge that this work is our own, and the use of Copilot 2026 (Microsoft, <https://copilot.microsoft.com>) and ChatGPT-5.2 (OpenAI, <https://chatgpt.com>) was done to summarise initial notes.

About Us

At engineering4confluence, we bring together diverse engineering expertise to deliver infrastructure that is resilient, sustainable and socially responsible. Our mission is to deliver solutions that strengthen communities, enhance the environment, and secure a sustainable future.

We approach every project through a truly holistic approach, ensuring that hydrology, ecology, geotechnics, water quality, carbon reduction and whole-life performance are considered from the outset. This enables us to design solutions that not only meet today's operational requirements but also anticipate the needs and pressures of future generations.

Our company ethics centre on collaboration. We engage proactively with stakeholders, regulators and local communities to ensure that our designs protect sensitive habitats, preserve cultural landscapes and enhance regional resilience. Sustainability underpins every decision we make, from low-carbon materials and nature-based solutions to safeguarding ecological flows and incorporating renewable energy technologies.

Through rigorous analysis, adaptive management and innovation, we strive to create water infrastructure that strengthens communities, respects natural systems and supports a secure, climate-resilient future.

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1.0 Introduction

Water is an essential aspect of everyday life, driving the economy, supporting communities, and sustaining a rich natural environment. However, the UK faces an increasing number of challenges in managing its water resources due to population growth and the increasing frequency of drought events due to a warming climate.

These pressures require long-term strategic interventions, including the development of large-scale water storage, aimed at increasing water reserves, and/or large-scale water transport connections to move water to areas that need an increase in water resources. Along with this, water quality also needs to be maintained as contaminated clean water reserves are at an increasing risk of algae growth and other pollutants due to climate change and environmental challenges.

1.1 A Water Resilient Future

The West Country of England is currently supplied by four major reservoirs: Chew Valley Lake, Blagdon Lake, Cheddar and Barrow, which provide a combined storage volume of approximately 38 billion litres (Bristol Water, 2026a). Demand modelling indicates that, without intervention, the region will require an additional 260 million litres per day by 2055 to maintain supply resilience (EA, 2025b).

In aid of increasing the water resource in the West Country, there are plans to construct a new reservoir, known as Cheddar 2, near Cheddar in Somerset, which will provide a potential storage capacity of 8.2 billion litres (WCWE, 2026). However, this alone will not meet long-term supply needs to support the 2055 targets. Additional, reliable water resources will be required to secure a safe and sustainable water resilient future for the region.

1.2 Proposed Solution

The proposed solution is the Glastonbury Reservoir, a 3 billion litre, pumped storage supplied by a new abstraction point on the River Brue. The scheme is supported by a draw-off tower, a zoned earth embankment structure, and controlled inflow and outflow systems that maintain both supply resilience and ecological protection. Water is transferred to the new Honeyhurst Water Treatment Works (WTW) via a 16 km rising main, powered by a dedicated pumping station. The design incorporates upstream catchment measures and in-reservoir water quality controls to minimise algal growth, ensuring a reliable, high-quality water source for the region.

1.3 Scope of Project

The scope of the project includes:

- selecting a suitable site within the Somerset Levels
- identifying viable and reliable water sources
- designing a storage reservoir and embankment system
- developing a water-quality management strategy with emphasis on algal control
- designing a reinforced-concrete draw-off tower
- designing a distribution pipeline to Honeyhurst WTW
- demonstrating that the system can reliably meet the required yield of 8 ML/day in June–July and 2 ML/day for the remainder of the year, with a reliability of 99%

1.4 Project Aims

This project aims to provide future water resilience to the South-West of England. The proposed system shall “Reliably increase the supply of water in the Bristol Water area by 8 million litres per day (June/July) and 2 ML/day (rest of the year) while ensuring good water quality and minimising environmental impact. This supply requirement must be met with a yearly reliability of 99%.”

2.0 Literature Review

2.1 Introduction

Precedent reservoir schemes in Somerset offer critical learning for geotechnical design, water quality management and hydrological behaviour. These insights reduce design risk, improve cost-effectiveness, and help anticipate environmental and planning constraints relevant to the development of a new water reservoir within the Somerset Levels.

Somerset contains a group of reservoirs, Cheddar, Barrow, Chew Valley, and Blagdon, which form the backbone of the public clean water supply to surrounding communities (Bristol Water, 2026a). These assets sit within or adjacent to nationally protected landscapes and habitats, requiring balanced management of water supply, environmental protection, and recreational use.

The proposed Cheddar 2 Reservoir seeks to meet future water supply demands by increasing drought resilience. Assessing the decisions made for this new proposal within Somerset will demonstrate new techniques and technologies for sustainable development. The scheme is designed to operate as a summer peak resource through environmentally sustainable abstraction, reducing pressure on sensitive river systems, and strengthening interconnections across the region, improving drought resilience (WCWE, 2026).

2.2 Existing Reservoirs in Somerset

2.2.1 Barrow Gurney Reservoirs (“Barrow Tanks”)

The three Barrow reservoirs, first opened in 1852, function as key treated-water storage for Bristol Water. They present a useful case study of embankment dam maintenance, with historic leakage leading to repeated investigations and remedial programmes (Soil Engineering, 2026). Management of these Victorian constructions to maintain their use for modern safety standards demonstrates how construction methods are adapted to sensitive environmental contexts (Halcrow Group, 2005).

2.2.2 Blagdon Lake

Blagdon Lake was constructed between 1898–1905 (Bristol Water, 2026b). The River Yeo was dammed in order to supply the growing city of Bristol. Engineering records describe deep excavation challenges to achieve a watertight foundation, reflecting the geotechnical uncertainty common to early reservoir schemes (Kerly, 2021).

Blagdon is a designated Site of Special Scientific Interest (SSSI), valued for biodiversity and recreational fishing. The original steam pumping station remains, as a Grade II* listed structure, showcasing the evolution of water supply infrastructure.

2.2.3 Cheddar Reservoir

Completed in 1937, the Cheddar Reservoir is a circular, concrete-lined bowl embanked above the surrounding ground filled by abstraction from the Cheddar Yeo (Bristol Water, 2021). It is a designated SSSI for biological interest, notably wintering waterfowl and regularly attracts scarce birds. It also holds community value through recreation, including birdwatching, sailing, and walking.

Historic records and photographic archives emphasise labour-intensive construction in the 1930s, using steam power, including temporary rail connections, and extensive manual labour (BBC, 2013). This provides a precedent for how large earthworks were historically staged in the Somerset levels and the development of a community around water infrastructure.

2.2.4 Chew Valley Lake

Chew Valley Lake was created by the damming of the River Chew and flooding farmland in the Chew Valley, and it opened in 1956. (Bristol Water, 2021) Archaeological excavations prior to flooding uncovered Roman and prehistoric artefacts (Bristol Water, 2022) .

Today it is a SSSI and Special Protection Area (SPA), with a national birdwatching centre with over 260 species recorded. It is a key example of a reservoir successfully integrating into a protected landscape. Chew Valley Lake has significant recreational value, including bird hides, water sports trials, and nature trails (Bristol Water, 2026c). It is a key component of Bristol Water's supply system, with water from Chew supplying Bristol, via Barrow treatment works. It is also used to regulate flows in the River Chew through releases that help maintain it during drought periods.

2.3 Proposed "Cheddar 2" Reservoir

The Cheddar 2 Reservoir has emerged from regional water resource planning as an in-region best value solution intended to address future supply deficits driven by climate change, population growth, and the need to reduce abstraction from environmentally sensitive rivers, the Rivers Exe and Otter, providing an increased protection status for chalk streams and other habitats (WCWE, 2026)

Cheddar 2 is designed to work alongside the existing Cheddar Reservoir by reusing the Bristol Water abstraction licences for Cheddar Springs and the River Axe, allowing the operator to prioritise maintaining storage in Cheddar 1 while diverting surplus inflows to fill the new Cheddar 2 impoundment (WCWE, 2022). The current plans are for a 9,000 ML reservoir located immediately south of the current basin, making efficient use of the established raw water infrastructure and the hydrological behaviour of the springs Axe system. Modelling undertaken shows that the system can reliably support a 36 ML/day deployable output during the critical two-month summer peak, as well as a 9 ML/day continuous "sweetening" flow through the remainder of the year to maintain treatment process stability and pipeline readiness.

Raw water abstracted to the new reservoir will be transferred to a purpose-built Honeyhurst WTW. The water quality risks of the blend necessitate a multi-barrier treatment process. To manage elevated algal risk, the treatment configuration ensures compatibility with the receiving networks' requirements.

Programme and delivery planning currently assumes that Cheddar 2 can be construction-ready by 2030 and in full beneficial operation by approximately 2035, subject to consents, where the project aims to deliver around 20 ML/day, during the drier six months of the year, enough for over 100,000 people.

2.4 Key Lessons from Precedence

Drawing from the existing and proposed reservoirs within the Somerset area, evidence shows:

- Historic leakage at Barrow (Halcrow Group, 2005) and deep cut-off requirements at Blagdon (Kerly, 2021) demonstrate the importance of early geotechnical investigation, potential seepage analysis, and robust design of embankment for a sustainable design life. Through modern construction materials and techniques, mitigations to reduce implications with ground conditions should be avoided in unexpected situations during construction.
- Water quality and ecological sensitivity within Cheddar, Chew, and Blagdon illustrate typical challenges for shallow reservoirs in agricultural catchments, including nutrients, algae, and pesticide inputs. Cheddar 2's treatment design reflects these known risks and can be adapted for other future projects in the region.
- Existing reservoirs show successful integration of wildlife protection and recreational value, providing a model for designing Cheddar 2's community interface. Following wider values for

sustainability, these aspects should be integrated from an early stage to achieve a greater benefit.

- Initiatives such as ‘Chew Valley Reconnected’ (Bathnes, 2024) highlight opportunities for reservoir projects to contribute to nature recovery networks, green infrastructure, and water quality improvements at the catchment scale.
- Cheddar 2 was constrained by abstraction caps on the Axe and Yeo, showing that future reservoir yield is determined primarily by available hydrology rather than reservoir size.

2.5 Conclusion

Somerset’s reservoir portfolio provides valuable precedent across engineering, environmental management, and community acceptance. Cheddar 2 aligns with regional resilience needs and builds upon lessons from historical schemes, particularly regarding ground conditions, ecological sensitivities, and public amenity design, however, several uncertainties remain. With appropriate mitigation and design considerations, future reservoir schemes have the potential to be highly sustainable projects that have a beneficial contribution to regional water security, alongside many other local aspects, including recreation and biodiversity benefits.

3.0 Site Analysis

3.1 Introduction

Selecting a suitable location for the proposed reservoir is a critical step, requiring consideration of numerous factors. Given the ecological sensitivity and extensive SSSI and SPA designations across the Somerset Levels, site selection requires careful balancing of hydrological suitability, environmental protection, constructability, and community impact. An appraisal was therefore carried out using a defined set of criteria to ensure that potential sites aligned with the project scope and supported the overall aims and objectives.

3.2 Appraisal

3.2.1 Criteria

To meet the aims and objectives of the project, the appraisal considered a holistic range of impacts associated with each potential reservoir location. Six equally weighted criteria were developed, each addressing a key aspect of site performance:

1. Water resource availability – Assessment of the proposed water abstraction point to include the likelihood of obtaining an EA abstraction permit, as per the South and West Somerset abstraction licensing strategy (EA, 2025a). Flow rates will also be assessed based on the measured catchment area for the abstraction location.
2. Constructability – Consideration of the distance from the proposed location of Honeyhurst WTW to minimise the pipeline material required. Given the flat topography of the Somerset Levels, the location will not have a large influence on pumping pressures, and in turn, the energy needed. Transport access and proximity to major roads and urban areas were also evaluated.
3. Water source quality – Review of existing water quality data for the selected site, with emphasis on nutrient content due to the risk of algae growth. Additional risks such as ecological impacts, chemical contaminants, and potential sewage spill sources are to be highlighted.
4. Social and economic impact - Preference given to areas of flat topography with clear, open land to reduce resident relocation. Consideration of community impacts, including

local agriculture, as well as potential flood-risk reduction benefits resulting from the reservoir.

5. Land designation - Avoid sites of notable designation, so as not to cause large environmental detriment to the local area. This includes avoiding SSSI and protected wetland habitats within the Somerset levels.
6. Existing geology – Evaluation of superficial deposits and underlying strata for suitability for embankment construction, including availability of clay for core material, and risk of compressible peat layers. The location would benefit from impermeable bedrock 5-20m below the surface, to reduce the magnitude of seepage mitigation.

3.2.2 Site Selection

Based on preliminary observations of aerial maps (Edina, 2026)(Defra, 2026a), four distinct sites were identified, as shown in Figure 1. Selection focused on areas that were relatively open, flat, and situated outside designated areas, sufficiently identified on the data sources used. The four appraisal sites each offer a different strategic opportunity. These sites were then compared using a structured appraisal process to determine the most appropriate location for the reservoir.



Figure 1 - Appraisal locations across the Somerset Levels

3.2.3 Scoring

Table 1 presents the appraisal process and the outcomes for the four sites. An equally weighted scoring system was applied across the six criteria, using a scale of 1-5:

- 1 - Critical issue
- 3 - Acceptable outcome
- 5 - Beneficial outcome

If a score of 1 were given for any criteria, that site would be excluded from consideration due to the associated critical constraint. Otherwise, the site with the highest total score would be selected. An equal weighting was chosen as it would show which site presented the most holistic benefit, where the objective was to eliminate unsuitable sites, based on the six critically different criteria.

During the appraisal process, the following assumptions were made:

- Precipitation across the Somerset Levels is the same and therefore the same at all sites. For the appraisal, the catchment area is the criterion that highlights the potential flow rates at the abstraction source.
- Floodwater will have no negative impact on the water quality that is supplied to the mains, due to sufficient treatment at the Honeyhurst WTW. Therefore, areas defined as high flood risk within the Somerset levels on flood map data (GOV.UK, 2026) may be preferred, as a new reservoir will help reduce the flood risk in the selected area. However, solutions within the reservoir will still be needed to manage the water quality within the reservoir.
- Cost-related impacts are not a factor, as it is a summed factor of all the criteria, where increased costs would be applied to unfavourable conditions.

Table 1 - Appraisal criteria and scoring

Appaisal sites	Site 1	Site 2	Site 3	Site 4
Location (approx.)	51°15'31"N 2°48'56"W	51°12'04"N 2°52'31"W	51°07'44"N 2°42'09"W	51°14'00"N 2°44'32"W
Site Description	South of proposed Cheddar 2 reservoir location.	West of Westhay Moor National Nature Reserve, protected wetland area.	South East of Glastonbury. Avoiding sites of the Glastonbury festival.	South of proposed Honeyhurst WTW location.
Strategic opportunity	Adjacent to the proposed Cheddar 2 site, with opportunities to integrate operations.	Large open land located downstream of the nature reserve, with an opportunity for ecological development.	High flood-risk area where storage is beneficial to reduce the flood risk of the Glastonbury area.	Close to proposed treatment works, to minimise pipeline requirements.
Appaisal Criteria	Score /5	Score /5	Score /5	Score /5
Water resource availability	Extraction from River Axe (upstream of the Cheddar reservoir) Restricted below Q30 - the River Axe is already abstracted for the Cheddar reservoir supply. approx. catchment area 76.03km ² . 1	Extraction from River Brue (upstream of River Bridge, TA7 8NW), Available above Q70. approx. catchment area 366.36 km ² . 4	Extraction from the River Brue (South of Glastonbury), over 2km upstream of the Wessex Water sewage treatment works, Available above Q70. approx. catchment area 176.98 km ² . 4	Extraction from River Axe (North of Theale) Restricted below Q30 - the River Axe is already abstracted for the Cheddar reservoir supply. approx. catchment area 32.03 km ² . 1
Constructability	Distance from proposed Honeyhurst WTW site = 4.85km Access via Cheddar road or Notting Hill way from A38, 2.8 km to Cheddar 3	Distance from proposed Honeyhurst WTW site = 10.00km Access via Harp road from the M5, 5 km to Wedmore, 3 km to other smaller villages (Mark, Blackford) 3	Distance from proposed Honeyhurst WTW site = 13.40km Access via A361 and A39 near to site, 1.5 km to outskirts of Glastonbury 2	Distance from proposed Honeyhurst WTW site = 1.45km Access away from main roads, 5.5 km to Cheddar and 6.5 km to Wells 4
Water source quality	slight change to natural conditions of dissolved oxygen – no indication of poor water quality (no major issues). 4	potential out spill from wessex water sewage treatment works (Glastonbury/Streets) - slight change to natural conditions. 1	near natural conditions of dissolved oxygen – less harm to aquatic life Large catchment area poses a risk of agricultural runoff - existing data suggests this does not influence normal water quality. 3	near natural conditions of dissolved oxygen and maganes – less harm to aquatic life. 5
Social and economic impacts	Flood risk Moderate. Land End Farm Camping is adjacent to the site and grazing fields. 3	Flood risk Low. Local agricultural land (river side farm, house farm). 2	Flood risk Moderate - High. No buildings in the surrounding area, potential grazing fields. 4	Flood risk High (potential for low areas). No buildings in the surrounding area, potential grazing fields. 4
Land designation	Improved grassland, Priority Habitat inventory – floodplain grazing marsh . 2	Neutral/Improved grassland and near areas of fen, marsh and swamp. 2	Neutral/Improved grassland, (Middle Drove / Watchwell Drove). 4	Neutral/Improved grassland and near area of fen, marsh and swamp. 3
Existing geology	Superficial geology - Tidal flat deposits-Clay, silt and sand. bedrock - Mercia Mudstone Group-Mudstone and halite-stone. 3	Superficial geology - Tidal flat deposits-Clay, silt and sand. Bedrock - Charmouth Mudstone Formation-Mudstone. 3	Superficial geology - Alluvium-Clay and silt. Bedrock - Black Ven Marl Member-Mudstone (mudstone and limestone interbonded) . 4	Superficial geology - Peat/Tidal flat deposits,Clay, silt, sand and gravel. Bedrock - Mercia Mudstone Group-Mudstone and halite-stone. 2
Appaisal scoring				
Total score /30	16	15	21	19

3.2.4 Scoring Outcomes

During the assessment, Site 1 and Site 4 received a score of 1 for Water Resource Availability. Both sites rely on abstraction from the River Axe, where existing licensing constraints associated with the Cheddar Reservoir and proposed Cheddar 2 scheme (WCWE, 2026) significantly limit the potential for new abstraction. These constraints, therefore, exclude Sites 1 and 4 from further consideration. Site 2 also received a score of 1 for Water Source Quality due to the risk of sewage affluent from the potential out spills from Wessex Water sewage treatment works, located upstream of Site 2. Given the scale and strategic nature of the proposed reservoir, this poses a critical risk, resulting in Site 2 being excluded.

Following the exclusion of these sites, Site 3 remained the only option without a critical score. Although not scoring perfect scores in all criteria, it is the best balanced site, achieving the highest total score (21/30). As such, Site 3 was selected as the preferred location for the proposed reservoir.

3.3 Selected Location

Site 3 is located south-east of Glastonbury and will abstract water from the River Brue. The large catchment area of the River Brue offers a more favourable hydrological yield potential than the River Axe. EA licensing records indicate lower constraint for strategic abstraction relative to the Axe and Cheddar Yeo systems (EA, 2025a). It also lies upstream of the Westhay Moor National Nature Reserve, which provides opportunities to enhance water security for the wetland ecosystem and support flows during drought conditions. Additionally, the ability to attenuate and store floodwater presents wider benefits for downstream communities from flood risk (GOV.UK, 2026).

From a land-use perspective, Site 3 comprises open grassland, likely used for rotational grazing. Given the prevalence of similar land types across the Somerset Levels, development at this location is unlikely to cause significant detriment to local agriculture or ecology. The presence of Middle Drove through the site provides convenient access for construction activities, however, diversion or realignment may be required as the reservoir design progresses.

Overall, Site 3 provides the most balanced and sustainable opportunity for delivering the scheme objectives while minimising environmental, social, and engineering constraints. For clarity, the selected site will be referred to as the Glastonbury Reservoir for the remainder of this report.

3.3.1 Potential Challenges and Mitigations

While Site 3 is the preferred location, several challenges have been identified that will require careful management during subsequent design stages. Key issues and proposed mitigation strategies include:

- Land Acquisition - Early engagement with landowners will be essential. A phased negotiation approach supported by fair compensation packages is recommended to secure access and maintain positive stakeholder relationships.
- Seasonal Disruption - Site 3 lies relatively close to the Glastonbury Festival site. Construction activities should therefore be scheduled outside peak festival periods, with close coordination with event organisers to manage traffic, access, and temporary works.
- Agricultural Runoff and Nutrient Management - Due to the size of the upstream catchment, nutrient loading presents a potential water-quality challenge. Options such as reedbeds, buffer strips, and catchment management measures should be incorporated to reduce inflows of sediment and agricultural runoff.
- Distance to Honeyhurst WTW - Although the distance to the Honeyhurst WTW is greater than for other sites, pipeline alignment optimisation and energy-efficient routing will mitigate operational pressures and costs.

4.0 Water Quality

4.1 Water Sources

Understanding the water quality of the inflows into Glastonbury Reservoir is important as it will influence the required treatment of the abstracted water source and any water quality management schemes that will need to be incorporated into the project. Water quality classification will follow the standards outlined by the Environmental Agency, under the Water Framework Directive (WFD) (EA, 2022). Water quality data has been taken from the EA surveys (EA, 2026) for catchment data, which directly influences the water quality of the River Brue. Precipitation is seen as having minimal risk of influencing water quality in Glastonbury Reservoir, as it provides a very small percentage of volume compared to the abstraction source.

For the proposed abstraction location, the catchment data for the River Brue is from Tootle Bridge to Clyse Hole Water Body (Water Body ID GB108052021182), which is predominantly upstream of the site. The catchment for the water quality assessment is shown in Figure 2.

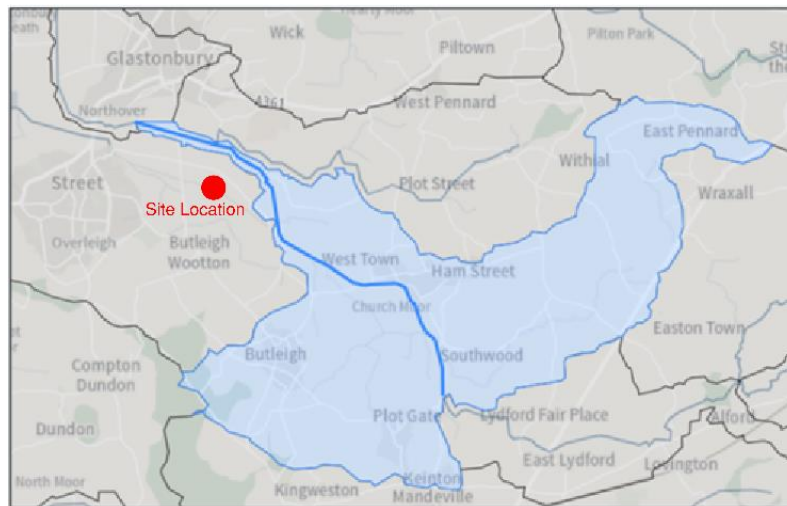


Figure 2 - River Brue and catchment from Tootle Bridge to Clyse Hole Water Body (EA, 2026)

4.1.1 Main Water Source

The main water source of the reservoir will be abstracted from the River Brue, with the latest water quality data collected in 2022 (EA, 2026). The River Brue is considered 'moderate ecological status', meaning that there has been a moderate change from its natural state as a result of human impact. This is likely due to nearby upstream farms, including Brook Farm, Ashview Farm, Tootle Bridge Farm and Lower Edgarley Farm.

The physical properties of temperature and flow rate for the River Brue were identified due to their influence on algae growth. Temperatures are currently near natural conditions, which helps maintain a high water quality. This is ideal as higher temperatures can increase algal growth rates. The River Brue observes higher than average flow rates, which help to promote mixing and aeration, increasing dissolved oxygen and can break up the formation of harmful algal blooms.

The chemical properties of the River Brue were identified as near natural conditions for dissolved oxygen, pH, zinc and copper, with moderate change to phosphate levels compared to natural conditions. Nitrogen and phosphorus measures were not available. A moderate change in phosphate levels may provide some additional nutrient input that can stimulate productivity for algae growth, so this will need to be monitored. Also, there is a risk of higher seasonal zinc and copper levels due to the upstream farms and fertilisers entering the river from runoff. Near natural levels of heavy metals reduce toxicity risk and help minimise the level of treatment required.

In terms of biological properties, the River Brue has normal levels of invertebrates and fish population, suggesting that conditions allow aquatic life to thrive, but fish safe measures will need to be taken into account when designing the water abstraction pumps during detailed design.

4.1.2 Run-off and Flood Risk

The risk of agricultural run-off and urban waste should be considered due to the upstream farmland and the proximity of both the River Brue and the proposed Glastonbury Reservoir to Glastonbury and surrounding roads. Flooding from the local area also poses a threat to water quality, as floodwater can mobilise contaminants, transporting them directly into the river system.

4.1.3 Groundwater

Groundwater is not likely to pose a risk to water quality, as the embankments will be raised above ground level and the reservoir will be impermeable, from the embankment foundation design.

4.2 Control Measures

Measures to improve water quality begin upstream of the abstraction point. A key mitigation involves active liaison with local farmers to ensure compliance with regulations such as The Reduction and Prevention of Agricultural Diffuse Pollution (England) Regulations 2018 (HMSO, 2018) and adherence to best-practice guidance such as the Nutrient Management Guide (RB209) (AHDB, 2017). The RB209 framework supports farmers in optimising the use of organic materials and balancing fertiliser application with environmental considerations, reducing the likelihood of nutrient run-off into the River Brue.

Continuous upstream monitoring will be implemented to assess water quality in real time and provide early warning of pollution events. This will allow abstraction to be paused immediately in the event of contamination, such as accidental spills or the introduction of hazardous pollutants into the watercourse. Monitoring data will also guide operational decisions and enable rapid response to emerging risks, supporting the long-term protection of the reservoir.

4.3 Further Data Collection

For certainty on future mitigation to improve the water quality in the River Brue, updated testing will be required, as the existing data is from 2022 (EA, 2026). During this testing, data will need to be collected on Phosphorus and Nitrogen content to inform the level of control for treatment and water quality management to ensure high-quality water is being abstracted into the Glastonbury Reservoir and mitigate algae development. Testing will need to be conducted throughout the year to assess seasonal variation.

5.0 Water Abstraction

5.1 Introduction

For the proposed Glastonbury Reservoir, water will be abstracted from the River Brue, South-East of Glastonbury, upstream of the Coxbridge brook (ST511372). The catchment area at the abstraction location is estimated to be 176.98 km², and is shown in Figure 3. To model the availability of water at the chosen location, a statistical flow model was created using existing measured data at measuring station 52010, Brue at Lovington (NRFA, 2025), 11km upstream of the chosen site, which captures 135.2 km² of the same catchment area, as shown in Figure 3. As this captures 76.4% of the same catchment, its measured flows can be reliably upscaled to represent conditions at the abstraction point. Because the River Brue catchment exhibits a relatively uniform hydrological response, upscaling is appropriate for this feasibility assessment.



Figure 3 - Catchment area of abstraction location and measuring station 52010

5.2 Statistical Flow Model

To characterise the variable behaviour of the monthly flow rates, a log-linear time-series regression model was used, as it ensures positive flows, which incorporates both seasonal effects and short-term temporal dependence. The model is expressed in Equation (1).

$$\log(\text{monthly rate}) = c_1 + c_2 \cos\left(2\pi \left[\frac{m - 0.5}{12}\right]\right) + c_3 \log(\text{rate previous month}) + \epsilon \quad (1)$$

Where m denotes the month of the year (January-December, 1-12), c_1 , c_2 and c_3 are model coefficients to be estimated, and ϵ represents a random error term.

5.2.1 Seasonal Effects

The seasonal component is modelled using a cosine term to capture the annual seasonality using a smooth periodic function with a 12-month period, where predictable cyclical patterns recur annually. The phase shift of 0.5 months centres the seasonal peak and trough within calendar months rather than at their boundaries, ensuring the fitted seasonal shape aligns more realistically with observed patterns.

The use of additional harmonic terms was considered to allow the model to capture multiple seasonal peaks or rapid fluctuations within a year. However, it was decided to maintain a simple model using only a single cosine term, as these patterns are not expected in this dataset, and including higher-order harmonics would increase the number of parameters without providing meaningful information. This results in only one coefficient being required for the seasonal effect, producing a more stable coefficient estimate that is less sensitive to noise in the data.

5.2.2 Lagged Dependent Variable

The inclusion of the lagged dependent variable reflects the natural persistence in river systems, where the flow in one month is strongly influenced by water stored in the catchment and hydrogeological conditions from the previous month.

5.3 Fitting Model to Measured Dataset

The dataset from measuring station 52010, Brue at Lovington (NRFA, 2025), provided daily measured flow rates from 1st October 1964 to 30th September 2024. Daily flow rates were summed into a monthly flow rate for use in the statistical model to reduce noise from short-duration storm events and align with typical monthly behaviours.

The log-transformed monthly flow, the seasonal cosine term and the lagged log flow were calculated to estimate the parameters c_1 , c_2 , c_3 and the standard error in Equation (1). The coefficients were obtained using the Regression module in Excel's Data Analysis ToolPak. The regression outputs are presented in Table 2.

Table 2 - Parameter outputs

	Intercept (c_1)	Cos Term (c_2)	Lag Log Flow (c_3)	Residual SD (σ)
Coefficient	0.885	0.271	0.450	0.258

After obtaining the coefficients, the model was used to generate predicted monthly flows across the same time period as the measured dataset. This enabled a direct comparison between the modelled and observed flow records to assess the reliability of the statistical model. An example of the measured flows and the modelled flows for a 10-year snapshot is shown in Figure 4.

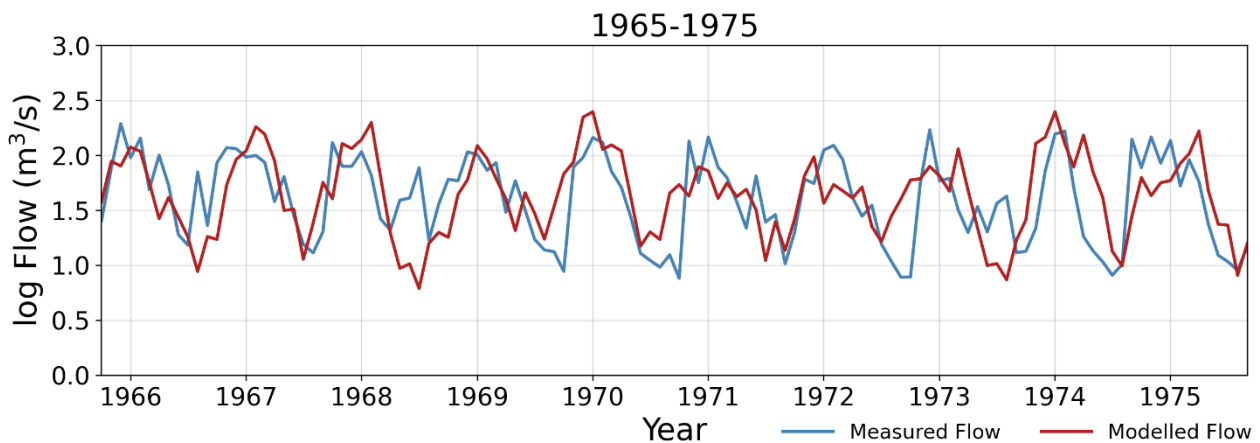


Figure 4 - Example flow data (1965-1975)

To reproduce the natural variability present in the historical dataset, the error term ϵ was incorporated in the simulation. This was implemented by multiplying the model's residual standard error by a standard normally distributed random variable. This ensured that each simulated month included an appropriate level of random hydrological variation around the deterministic model output.

Because the normal distribution occasionally produces extreme values that are not realistic for river systems, the random term was bounded within ± 2.5 standard deviations. These limits were set to remove the extreme outliers that occur less than $\sim 1\%$ of the time, known as black swan events, as these scenarios are unrealistic and if they were to occur, the river would be flooded along with the surrounding areas, making this model redundant.

By adjusting the magnitude of the error term within the bounds of the residual variation, the simulated flows were matched more closely to the spread and behaviour of the measured dataset, providing a clearer indication of the model's reliability. This calibration confirmed that the fitted model successfully reproduced both the seasonal behaviour and the month-to-month persistence evident in the historical flow record. Figure 5 compares the monthly probability density functions (PDFs) of the measured and modelled flows, demonstrating the model's ability to replicate the seasonal variability in the River Brue.

The normal root mean squared error was found to be 0.20 for the comparison between the observed data and the fitted modelled data. Statistically, this falls within the range typically interpreted as good agreement in engineering and environmental modelling applications, indicating that the model replicates the observed distribution with acceptable precision.

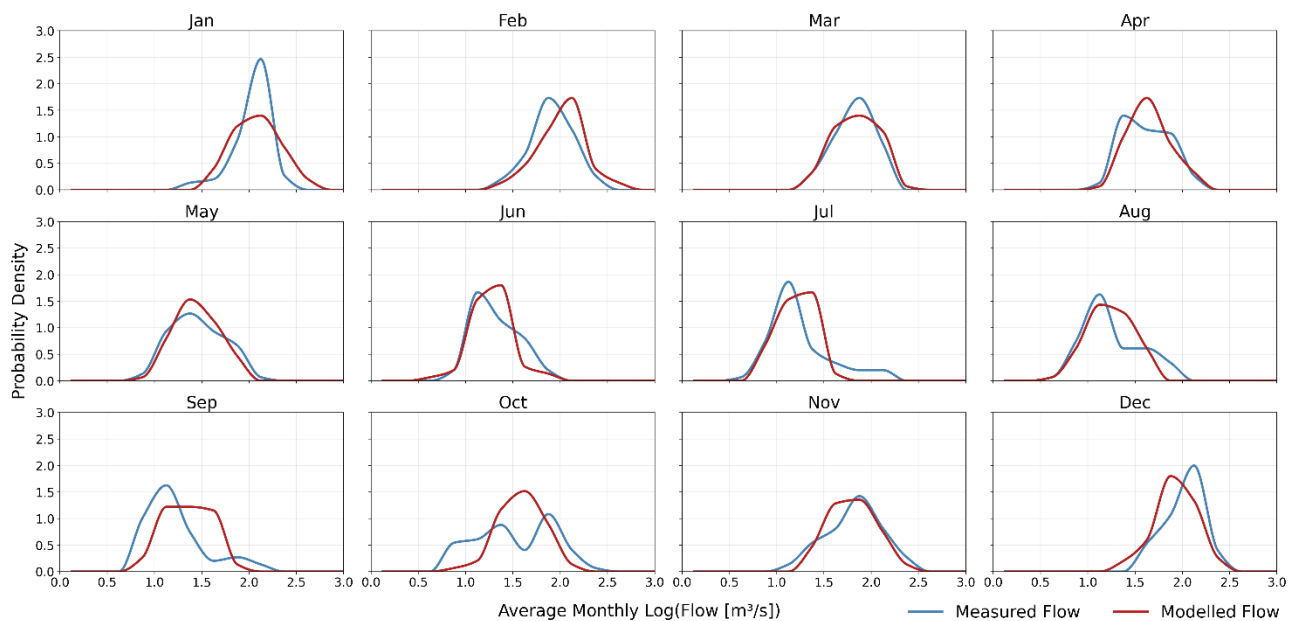


Figure 5 - PDF comparison

5.4 Rate of Abstraction

Using the fitted statistical model, and an upscaling factor of 1.3 for the increase in catchment size of the desired abstraction point, a continuous series of simulated monthly flows was produced for the River Brue at the proposed abstraction point to model the flows from the year 2000 to 3000. This allowed the long-term flow behaviour at the site to be assessed and used to inform a suitable and sustainable abstraction rate for the Glastonbury Reservoir supply.

A key design metric for abstraction is the Q70 flow, defined as the discharge that is exceeded, on average, 70% of the time. This represents an estimate of available flow during typical hydrological conditions, ensuring that abstraction does not cause low-flow periods or significantly affect ecological activity.

From the simulated flow record, an average Q70 value for the River Brue at the abstraction location was determined. This value was then compared with the proposed maximum daily abstraction volume for the reservoir of 50 ML/d.

By assessing the simulated monthly flows relative to the Q70 threshold, it was determined that the proportion of flow in an average year is surplus in supporting the scoped demands of the proposed project outputs of 8 ML/day (June/July) and 2 ML/day (rest of the year). This provides a robust basis for evaluating sustainable abstraction limits at the proposed site and the abstraction is not limited by the scope but instead the optimisation for abstraction rates that will provide a resilient water supply, with sustainable design decisions taken to support economic and environmental factors.

6.0 Reservoir Design

6.1 Introduction

From the flow model developed, it was determined that the River Brue has a surplus supply of water for the minimal scope of the project. Due to this finding, the opportunity was taken to optimise the intake and output of the reservoir to develop the sizing of the reservoir, so as to allow for ecological benefits, future water resilience and flood alleviation for downstream areas. From this, the embankments were designed to ensure long-term durability.

6.2 Reservoir Sizing

Modelled flow data for a 1000 year period was used to develop the size of the proposed Glastonbury Reservoir. The following constraints were used to ensure the design met the required scope:

- Supply the Bristol Water area with a minimum of 8 ML/day (June/July) and 2 ML/day (rest of the year). Bristol Water quote that the average household they supply uses 356 L/day (Bristol Water, 2025a), leading to a minimum of roughly 5,618 households supplied daily.
- The River Brue can only be abstracted from 70% of the time, with EA restrictions of Q70 flow.
- The maximum rate of abstraction is 50ML/day.
- The supply requirement must be met with a yearly reliability of 99%, where a minimum of 5% of the total capacity (not including free board) is maintained in the reservoir to ensure the ecology is not lost.

This baseline option was modelled, with maximum abstraction rates and the minimum required outputs. Due to the high flows in the River Brue, it was found that 900,000 m³ was the minimum volume required to meet the scope. However, this is an inefficient use of resources and steps were taken to develop additional benefits to take advantage of the positioning and availability of water to achieve a sustainable and future resilient option. From this, a suitable reservoir size was determined.

6.2.1 Increased Output Supply

It is expected that the population of Bristol will increase by 50,000 people by 2050 (ons, 2022), approximately 12,500 households. To design a truly resilient water storage system, the surplus water supply provided by the River Brue allows for an increased baseline supply. A maximum supply increase to 20 ML/day in June and July and 5 ML/day for the rest of the year, to supply a minimum of roughly 14,000 households daily, is to be achieved. This increase assumes that the capacity at Honeyhurst WTW will be sufficient to handle this, or that the capacity for water treatment will be increased when needed.

6.2.2 Optimised Inflow Pumps

For a project aiming for a design life of 150 years, a set of pumps with a total capacity of 50 ML/day will be installed for future resilience of the site. However, the pumps should not run at full capacity, the additional capacity provides redundancy for repairs and potential use in extreme events to alleviate flows in the River Brue. For the model, it is assumed that a maximum intake of 20 ML/day is abstracted whenever River Brue flows are at or above the Q70 threshold.

6.2.3 Flood Alleviation

Due to the high flood risk in the area where the proposed construction of the Glastonbury Reservoir is, it is proposed that the reservoir maintain an allowance for flood capacity to reduce the risk of flooding of the River Brue, downstream of the reservoir.

For the modelled flow, flood events are considered for the 95th percentile of flows. These high flow rates are associated with rising levels within the River Brue. With Glastonbury Reservoir constructed, additional water will be abstracted from the River Brue to reduce the height of the water level in the river, minimising the effects of potential flood events. Although in extreme events, like a 1 in 200 year flood event, it is impossible to fully negate the impacts of the flooding caused.

In the modelled flood events, it is simulated that the intake from the River Brue is increased to 40 ML/day, from 20 ML/day, to reduce the flows in the river at a greater rate. To allow for this capacity, the reservoir will be modelled to maintain a maximum capacity of 75% of its total volume to yield an allowable surge storage capacity, which will fill to 100% capacity during flood events, whilst maintaining an adequate supply for the summer months.

6.2.4 Drought Support for the Nature Reserve

Downstream of the proposed reservoir is the Westhay Moor National Nature Reserve and other protected wetland habitats where water from the River Brue helps to maintain the complex habitats. As a part of the storage of water upstream of these areas, it is proposed that during periods of low flow, when the wetlands may experience potential drought, the reservoir can discharge a flow of water back into the River Brue to maintain the ecology.

For the model, if flows are below the 5th percentile within the River Brue, it is modelled that a flow of 2 ML/day is discharged back into the River Brue to maintain a good ecological condition. This will reduce the failure of good ecological status by maintaining flows above the threshold environmental flow (Defra, 2026b).

6.2.5 Final Size

A volume of 3,000,000 m³ is outlined as the total required capacity from the model as a result of the additional conditions above. This was determined through the set measures and adjusting the volume to find a size which produced a consistent 99% reliability from the modelled flow, over a 1000 year period. With this design volume, it is modelled that the reservoir will be unable to supply the full 20ML/day for approximately a 1% annual probability, but a portion of that will still be made available. Figure 6 shows a 100 year snapshot of how the volume in the reservoir changes over time.

This final sizing does not take into account gains from precipitation or losses from evaporation as it is assumed that these are negligible compared to the pumped inputs and outputs. The model also doesn't take into account the potential volume due to freeboard, as this is not to be actively used as storage.

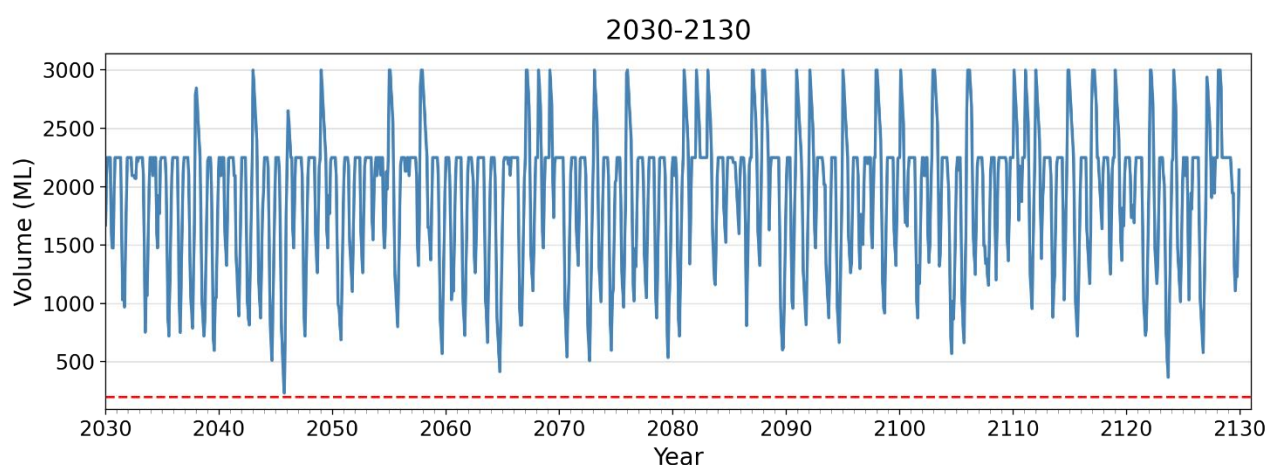


Figure 6 - Example modelled reservoir volume (2030-2130)

6.3 Reservoir Shape

The geometry of the reservoir plays a significant role in mitigating algal growth and maintaining overall water quality. Deep, narrow reservoirs are particularly effective due to their hydrodynamic and thermal characteristics, and the incorporation of curved corners further reduces the risk of stagnation zones forming. The proposed shape of the Glastonbury reservoir is shown in Figure 7.

A deep water profile limits light penetration, thereby suppressing photosynthetic activity and reducing the potential for algal growth. The reduced surface-area-to-volume ratio associated with narrow reservoirs also minimises solar heating, helping to prevent thermal stratification that can create favourable conditions for algal blooms. Additionally, the elongated plan shape promotes more turbulent flow conditions by increasing the fetch distance in the dominant wind direction, this also

discourages the development of stagnant areas. By enabling a more direct through-flow from inlet to outlet, the overall residence time is also reduced. The final reservoir dimensions, as shown in Figure 7, were directly influenced by the requirements of the embankment design, ensuring that hydraulic performance and structural stability are achieved concurrently.

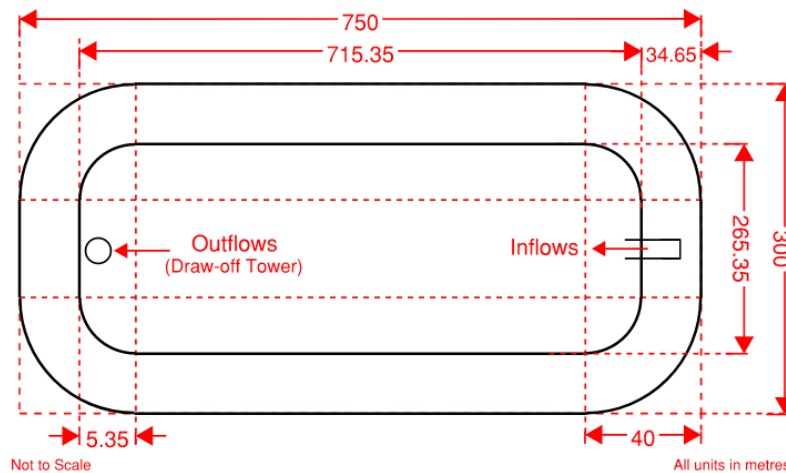


Figure 7 - Glastonbury Reservoir shape and size

6.4 Embankment Design

The construction of the Glastonbury Reservoir requires the formation of engineered earth embankments on all sides, as the reservoir is located on a flat, low-lying floodplain with no natural topographic containment. The reservoir is therefore fully formed through excavation and the embankments. A comprehensive preliminary assessment was undertaken to establish the ground conditions. The embankment geometry was then optimised using numerical stability modelling under a range of operational and extreme loading scenarios, with the resulting design meeting the performance requirements.

6.4.1 Geological and Geohydrological Data

Borehole data directly at the proposed site are not available. The nearest publicly accessible records are located approximately 2.5 km west of the site (boreholes ST43NE1 and ST43NE2) and 3.2 km southeast (borehole ST53SW1), as collected from British Geological Survey data (BGS, 2026a). Figure 8 displays the locations of the boreholes in relation to the site boundary, alongside the expected geological conditions obtained from BGS data (BGS, 2026b). All three were drilled in 1972, and no more recent publicly available borehole data exists for the River Brue floodplain corridor.

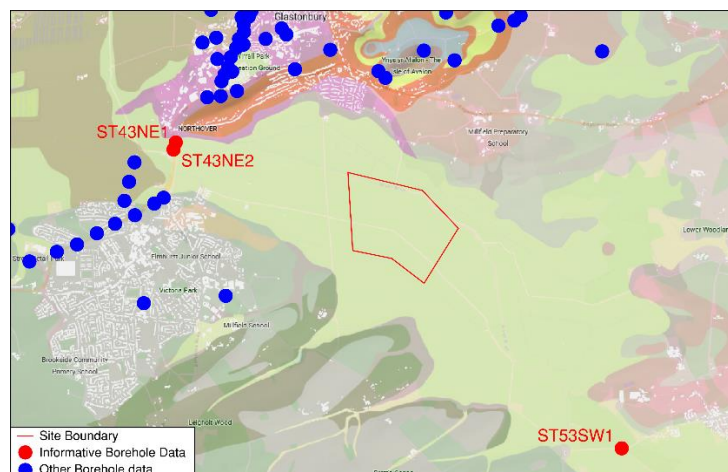


Figure 8 - Borehole location plotted on BGS geological map (BGS, 2026b)

Comparison of these logs with regional geological mapping indicates that ST43NE1 and ST43NE2 are not representative of the expected current site conditions. These boreholes lie close to the extensive peat-rich wetlands west of Glastonbury, where superficial deposits are highly variable and locally deep. The borehole ST53SW1, however, lies within a more typical alluvial setting and was therefore considered a more suitable representation for the geological profile of the site. The interpreted geological model for the proposed site location is shown in Figure 9, alongside boreholes ST43NE2 and ST53SW1.

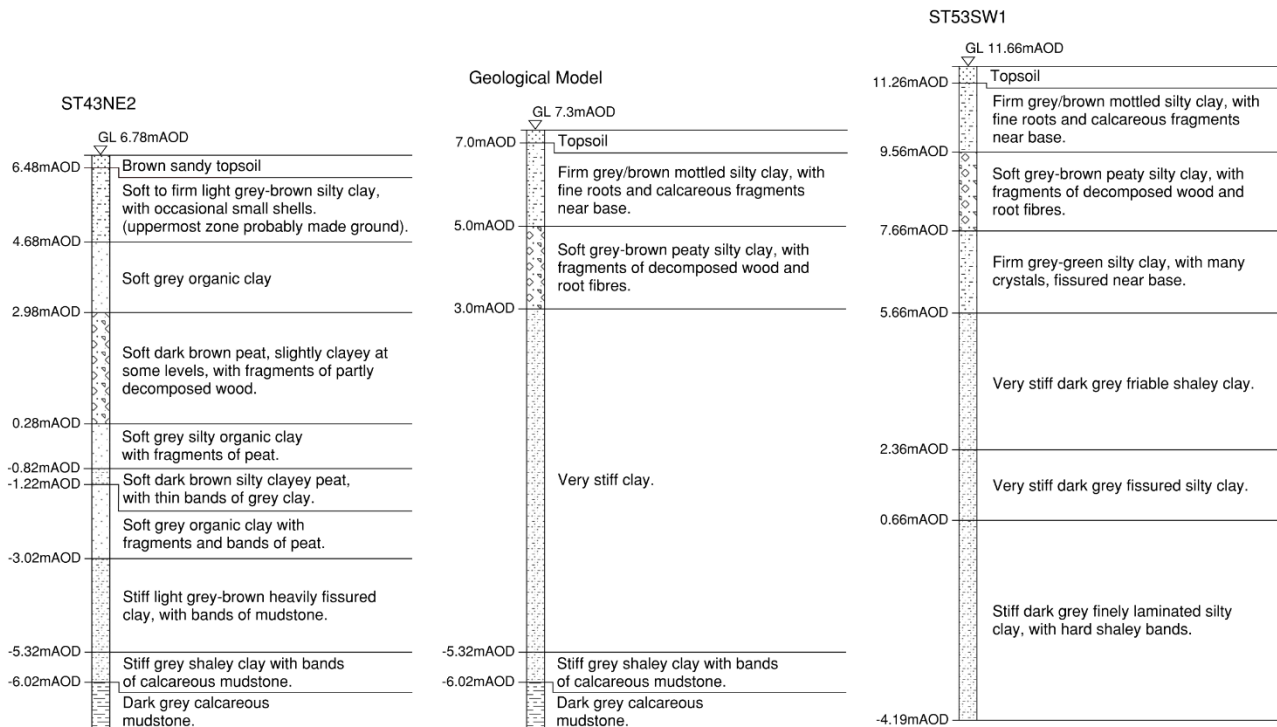


Figure 9 - Borehole data and geological model

Although this approach relies on historical and off-site data, it is a reasonable assumption for the initial design stage, because the recent British Geological Survey data confirms the similarities of the historic measured data. A site-specific ground investigation will, however, be required to confirm material variability, groundwater conditions, and engineering properties for detailed design.

The site itself is topographically flat, with ground elevation around 7.3mAOd $\pm$ 0.5 m. Superficial deposits at the site are interpreted to extend to approximately 3.0mAOd, comprising alluvial clay and silt with organic inclusions. These sedimentary deposits are fluvial in origin, reflecting the deposits into the floodplains of the River Brue. Below 3.0mAOd, the model indicates very stiff clay. Bedrock is found at a -6.00mAOd and is classified as Black Ven Marl Member (BGS, 2026b), an impermeable bedrock composed of fine-grained, dense mudstone.

The groundwater level was not referenced on the historic borehole data. Therefore, for modelling purposes, and consistency with the Somerset Levels, the groundwater table has been assumed to be at ground level, which is compatible with a shallow regional water table in floodplain environments. This assumption is conservative, with fully saturated ground conditions.

6.4.2 Design Considerations

As the reservoir is not impounded on a natural watercourse, no spillway is required. All inflows are fully controlled via the intake pipes and pumping system, and the reservoir is therefore not exposed to uncontrolled catchment-derived flood inflows. Direct rainfall onto the reservoir surface is limited and can be safely accommodated within the available freeboard allowance.

A zoned earth embankment incorporating a central clay core has been selected for the design. The clay core provides the primary impermeable barrier, preventing seepage through the dam body. This zoned arrangement is preferable to a homogeneous embankment, which, although simpler to construct, is historically associated with an increased risk of excessive seepage and cracking, as it uses a mixed mass of soil. On the clay core, a granular outer shell will be constructed to provide structural stability and is more economical and quicker to compact, while the clay core, although slower and costlier to place, ensures an effective hydraulic cutoff. This hybrid design is vital for safety and sustainable economic construction. Clay used in the core must be compacted at its optimum moisture content, typically between 12–18%, to achieve the required impermeability and strength.

An important failure mechanism considered in the design is piping, defined as the internal erosion of an embankment caused by water seeping through and carrying soil particles with it. The zoned earth embankment, with its impermeable clay core supported by a granular shell, is essential in reducing this risk by both limiting seepage and providing sufficient resistance against internal erosion.

Ground conditions beneath the proposed embankment comprise approximately 8 m of very stiff clay. However, historical borehole records indicate variability within this layer, including patches of fissured clay containing natural cracks and joints. This increases the likelihood of seepage paths forming beneath the embankment foundation, potentially leading to piping and progressive erosion. To mitigate this risk in the initial design, it is assumed that sheet piling will be required. The sheet piles would extend from the embankment core down to the underlying bedrock, creating a continuous impermeable barrier that prevents seepage. Following completion of a detailed site investigation and confirmation of the actual ground conditions, this requirement can be re-assessed, and the sheet piling may be omitted if shown to be unnecessary, providing a potential cost saving.

6.4.3 Modelling Design

The slope stability of the embankment was assessed using the HYRCAN modelling software, which calculates the factors of safety (FoS) using a simplified Bishop's method. This assessment method is good for this early-stage proposal, but a more detailed geotechnical analysis will need to be conducted in the detailed design. When developing the model, topsoil was not included in the model as it is assumed to have no structural properties. Bedrock is also not included in the model, as it is assumed to be a solid, impermeable layer.

To fully assess the stability of the embankment, it was evaluated under a range of critical loading and reservoir conditions, including seasonal water level fluctuations, steady state seepage, rapid drawdown, and flood loading conditions. Several operational scenarios were modelled to represent the anticipated annual variation in reservoir water levels, as summarised in Table 3. In addition to the normal operating conditions, a special external flooding scenario was also evaluated to assess the effects of surrounding floodwater on pore pressure and stability on the downstream side of embankments. The reservoir water level is shown to the left of the embankment in Figure 10 and the external water level on the right.

Table 3 - Reservoir and external water levels

Category	Condition	Water Level	
		Relative to reservoir base (m)	mAOD
Reservoir	Drought	1	4
	Summer	4.875	7.875
	Full	11.25	14.25
	Flood	15	18
External	Drain	0	3
	Flood	4	7

Soil parameters were modelled using both a base case and a worst case condition, representing the expected range of properties for the materials present within the embankment and foundation. These parameters are presented in Table 4. In the worst-case condition, a conservative assumption of zero effective cohesion was adopted, reflecting the fact that long-term drained behaviour and fluctuating reservoir levels do not provide reliable sustained cohesion in clay soils. This approach also ensures that the analysis remains conservative under both steady-state seepage conditions and scenarios where pore pressures have not fully dissipated following changes in water level.

Table 4 - Soil properties for Full Drain conditions

No.	Soil Type	Cohesion, C (KN/m ²)			Friction Angle, ϕ (°)			Unit Weight, γ (KN/m ³)		
		Base case	Worse case	range	Base case	Worse case	range	Base case	Worse case	range
1	Clay Core	30	0	0-30	15	12	12-15	21	21	20-21
2	Sand Filler	0	0	0	36	34	34-36	20	20	19.5-20
3	Shell ¹	0	0	0	35	32	32-38	22	23	22-23
4	Firm silty clay	0	0	0	26	24	24-28	20	20	19.5-20
5	Soft peaty silty Clay	15	0	0-15	24	22	22-26	17	17.5	16.5-17.5
6	Very Stiff Clay	20	0	0-20	31	29	29-33	21	21	20-21
7	Clay Filler	20	0	0-20	31	29	29-33	21	21	20-21

¹Shell is silty sandy Gravel with fragments of crushed rock

The embankment geometry was progressively refined to achieve the target FoS agreed with the client for static slope stability of earth embankments, in the range between 1.3 and 1.5 for long-term operational conditions. The FoS results of each modelling scenario are summarised in Table 5, and an example of the HYRCAN model outputs is shown in Figure 10. For Scenario 10, the FoS was calculated as 1.243, which falls below the target value of 1.3. However, this was considered acceptable due to the very low probability of occurrence, as the scenario represents a persistent 1-in-100-year flood event sustained over a two-month rolling period.

Table 5 - HYRCAN model results

Scenario	Soil Properties	Water level		FoS in Failure Direction	
		Reservoir	External	Right to Left	Left to Right
1	Base	Drought	Drain	1.471	1.437
2	Base	Summer	Drain	1.472	1.437
3	Base	Full	Drain	1.499	1.437
4	Base	Flood	Drain	1.474	1.437
5	Base	Flood	Flood	1.474	1.493
6	Worse	Drought	Drain	1.313	1.313
7	Worse	Summer	Drain	1.316	1.313
8	Worse	Full	Drain	1.336	1.313
9	Worse	Flood	Drain	1.315	1.313
10	Worse	Flood	Flood	1.315	1.243

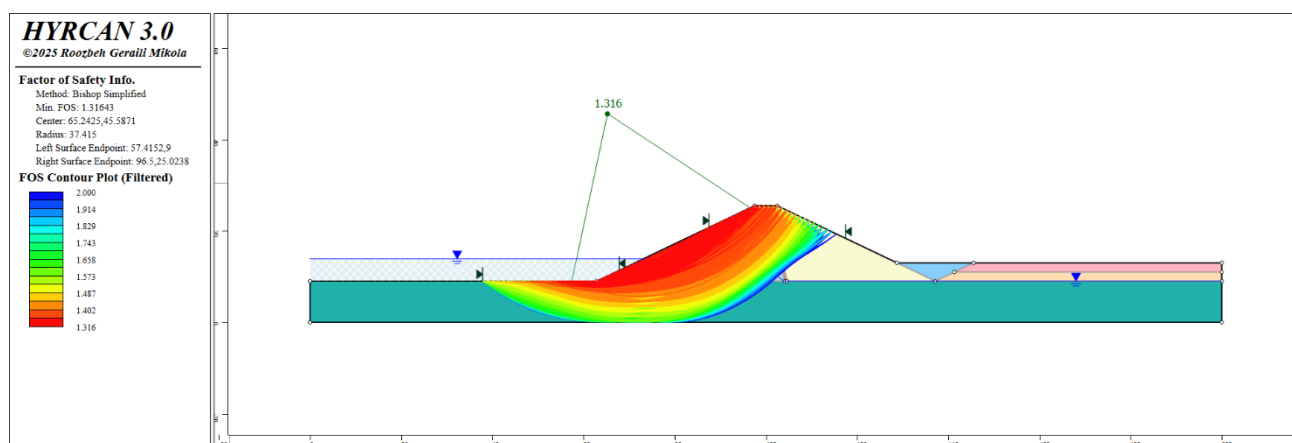


Figure 10 - HYRCAN model output (scenario 7 – Right to Left)

6.4.4 Final Design

Figure 11 illustrates the final embankment geometry adopted following completion of the stability assessment. All dimensions used in the HYRCAN analysis are labelled directly on the figure.

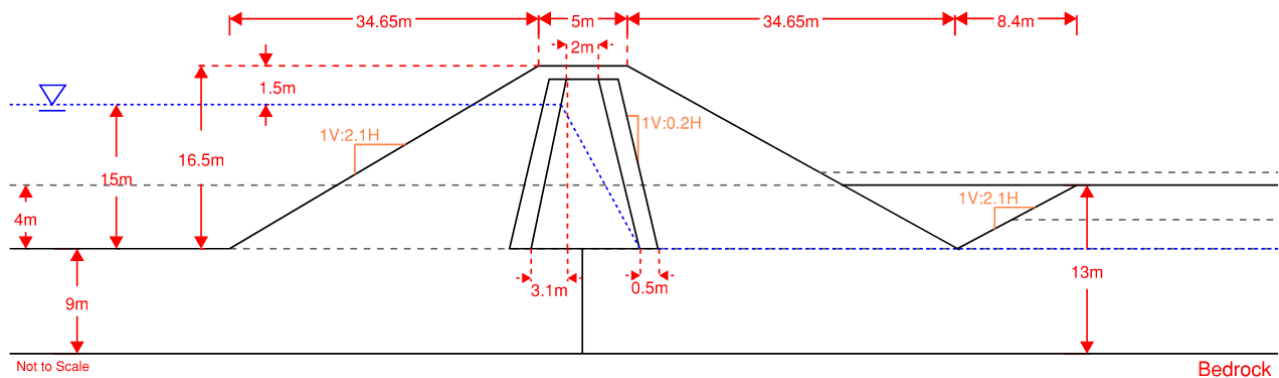


Figure 11 - Embankment design

The total embankment height from foundation level to crest is approximately 16.5 m on the upstream side, with the reservoir water depth designed to a maximum of 15 m. A 1.5 m freeboard allowance has been incorporated above the normal maximum reservoir level. This provides adequate protection against wave action, settlement, and operational uncertainties, ensuring that overtopping risks remain acceptably low even under extreme water level fluctuations.

The embankment has been designed with a 5 m wide crest, providing sufficient width for safe vehicle access during operation and maintenance while also allowing for future inspection works. Beneath the crest, the central clay core has a minimum width of 2m, ensuring an impermeable barrier throughout the height of the embankment. On each side of the clay core, a 0.5 m sand filter layer is provided. These filters act as transition zones, allowing controlled drainage while preventing the migration of fine clay particles, thereby reducing the risk of internal erosion and ensuring the long-term integrity of the core.

The upstream and downstream embankment shells adopt 1V:2.1H side slopes, providing a stable geometry that satisfies the required factors of safety under all operational and extreme loading scenarios. The downstream side also incorporates a secondary drainage bench positioned above the external water level for the flood scenario, improving pore pressure dissipation and enhancing stability.

Overall, the final embankment geometry satisfies the design criteria for a zoned earth embankment, providing adequate stability, impermeability, and freeboard capacity while remaining compatible with the geotechnical uncertainties identified during preliminary investigation.

6.5 Construction

Construction of the embankment requires the removal of approximately 4.5 m of material to remove existing silty clay, which is unsuitable for the foundation of the embankment structure. The total excavation volume is estimated at 296,000 m³, all of which will be processed through an on-site material management plant. This approach minimises the need for off-site disposal and reduces the volume of imported materials. Preliminary assessments indicate that the excavated material can be effectively separated to recover sufficient clay of suitable quality for use within the central clay core, supporting a sustainable and cost-efficient construction strategy.

The embankment will be constructed in controlled 30 cm thick layers, compacted to achieve the specified density and moisture content. This method ensures the clay core develops the necessary

low-permeability structure, while the granular shoulders achieve the strength required to maintain long-term stability. Quality control testing will be undertaken throughout construction, including in-situ density tests and moisture content monitoring.

Given that the embankment is founded on very stiff clay, differential settlement is expected to be minimal. The uniformity and stiffness of the founding layer help to ensure that any settlement which does occur will be broadly uniform across the width of the embankment, avoiding distortion of the clay core. Nonetheless, a staged construction approach and ongoing monitoring will be implemented to verify performance as fill is placed and compacted.

7.0 Managing Reservoir Water Quality

7.1 Introduction

Maintaining high water quality within the reservoir is essential to ensuring a reliable drinking water supply and minimising treatment requirements at Honeyhurst WTW. As outlined in Section 4, the primary inflow to the reservoir will be abstraction from the River Brue, which is currently classified as having 'moderate ecological status' (EA, 2026). Upstream agricultural activity introduces risks associated with nutrient loading from organic matter, which can increase algal growth.

Physical, chemical and biological processes all interact to determine the overall water quality. Nitrogen and phosphorus from agricultural runoff are of key concern as they promote biological productivity and often lead to large algae blooms, particularly during periods of warm temperatures. If unmanaged, excessive algal growth can result in eutrophication, reducing dissolved oxygen concentrations and degrading aquatic ecosystems.

To prevent eutrophication, Glastonbury Reservoir will be managed with the objective of maintaining the mesotrophic trophic class, characterised by clear water with beds of submerged aquatic plants and medium levels of nutrients and oxygen, with a low algal potential. Achieving this condition requires a holistic approach, combining reservoir design, in-reservoir mitigation strategies and catchment-scale management. The reservoir's narrow, deep geometry also supports water-quality management by limiting light penetration and reducing thermal stratification. The following sections outline the proposed measures to restrict algae growth, maintain good water quality and ensure the long-term resilience and sustainability of Glastonbury Reservoir as a reliable drinking water source.

7.2 Water Treatment

Experience from comparable reservoir schemes within the region indicates that algae growth represents one of the most significant risks to water quality. For example, the proposed Cheddar 2 reservoir identified several water quality risks associated with inflows from the River Axe, including increased algal potential due to variable abstraction points and the presence of pesticides within the catchment.

To mitigate such risks, the Honeyhurst WTW is currently designed to include (WCWE, 2022):

- DAF clarification - removes suspended solids, oils and grease
- Rapid gravity filtration - removes suspended solids, turbidity and algae
- GAC adsorbers - remove organic chemicals, micropollutants and VOCs
- UV and chlorine disinfection - kills pathogens (bacteria and viruses)
- Orthophosphate dosing - for corrosion control in WSX networks

Although these treatment processes provide effective downstream protection, it remains preferable to minimise water quality problems within the reservoir itself. This minimises the load on the treatment plant, which can reduce costs, improve operational resilience and enhance the ecological and biodiversity health of the reservoir and surrounding area.

7.3 Restricting Algae Growth

The proposed algae-management strategy for Glastonbury Reservoir combines mechanical mixing, nature-based solutions, targeted monitoring and contingency measures for algal removal. Together, these approaches aim to maintain stable water quality, reduce treatment demand and support a resilient ecological system within the reservoir.

7.3.1 Bubble Plumes

Reservoirs often develop stratification, where the water column separates into layers of differing density. This prevents vertical mixing and limits the exchange of heat, dissolved oxygen and nutrients. The upper epilimnion is typically warmer and oxygen-rich due to direct solar heating and atmospheric contact, allowing for oxygen to diffuse directly from the air. In contrast, the deeper hypolimnion becomes isolated and oxygen poor due to oxygen consumption from the decomposition of organic matter settled on the reservoir bed. These low oxygen conditions can threaten aquatic ecosystems and promote the release of soluble iron and manganese from sediments (Birt et al., 2021).

To mitigate these water quality risks in the proposed Glastonbury Reservoir, a bubble plume destratification system is recommended. Bubble plumes introduce compressed air through diffusers located at the reservoir bed. As the bubbles rise, they entrain water and create upward circulation, which destratifies the water column and more evenly distributes oxygen throughout the reservoir (Choi and Kim, 2024). This mixing improves both biological and chemical water quality and reduces the likelihood of harmful algal blooms, including cyanobacteria, which thrive under stable, stratified, warm surface-water conditions.

7.3.2 Nature-Based Solutions

In parallel to mechanical mixing systems, the Glastonbury Reservoir will incorporate nature-based solutions designed to improve water quality and enhance ecology. These solutions primarily work by harnessing natural processes to filter nutrients and reduce sediment loads. Nature-based approaches have several advantages, they can improve water quality at a relatively low cost, create new habitats and increase resilience to future environmental changes.

7.3.2.1 Floating Wetlands

Floating wetlands will be incorporated to support nutrient removal and biodiversity enhancement. This system consists of buoyant platforms supporting vegetation, with plant roots hanging down into the reservoir. Roots absorb nutrients such as nitrogen and phosphorus, limiting nutrient availability and reducing the risk of algal growth. Additionally, microorganisms that grow on the plant roots further increase nutrient removal through biological processes (Wang and Sample, 2014). In addition, the wetlands increase biodiversity by providing habitats for fish, birds and other wildlife.

Different plant species are known to have different effectiveness in pollutant uptake. The proposed design includes common reed, yellow flag iris, jointed rush and broadleaf cattail, selected for their high biomass production, as it is known that a wetland containing a diversity of species has a more effective uptake. These species also have the benefit of reducing a large range of pollutants, ammonia, nutrients, heavy metals, oil and grease, and E.coli. Emerging evidence indicates that floating wetlands can also remove emerging contaminants such as polyfluoroalkyl substances, which are not always filtered at treatment plants (Awad et al., 2022). However, cover of more than 50% of the reservoir can cause anoxia, and a low cover of 9-18% may provide little treatment effect. For Glastonbury Reservoir, a recommended cover of approximately 30% of floating wetlands will allow for effective nutrient uptake and accessible maintenance of the plants, without risking anoxia.

Active management is required to maintain performance. If vegetation is not periodically harvested, nutrients stored in plant biomass may be released back to the reservoir as organic-rich sediment. Some of these minerals, such as phosphorus, can become biologically available and return to the water, and once again lead to algae blooms. Harvesting plant shoots has been shown to be the most practical and cost-effective approach, with rapid regrowth from retained root systems.

There is further opportunity to improve the nutrient uptake of the floating wetlands. Innovative technology, trialled by Bristol Water at Chew Valley Lake, involved the use of solar-powered pumps to pull water up from the lake and deposit it on the clay bags, which filter and bind unwelcome nutrients (Bristol Water, 2025b). Once these clay bags are full, they can be replaced and reused by local farmers to help grow crops, creating a circular nutrient management system within the catchment. This will promote knowledge sharing to help develop this innovative scheme and aid relations with local farmers in promoting agricultural development in line with regulations.

7.3.2.2 Constructed Wetlands Surrounding Reservoir

Constructed wetlands and reedbeds will be established around the perimeter adjacent to the River Brue, to provide an additional layer of water quality management (Ferreira et al, 2023). These wetlands intercept and filter embankment drainage, functioning similarly to floating wetlands by removing nutrients and suspended solids. The resulting blue-green landscape enhances biodiversity, supports birds and wildlife, and offers recreational benefits. Additionally, when the flow in the Brue is low, characterised as being below the Q5 threshold, stored reservoir water will be returned to the river via these constructed wetlands, supporting ecological flow and ensuring water is released in a treated, low-nutrient state.

7.3.3 Monitoring

Monitoring is a critical component of effective reservoir water quality management. As mentioned in section 4, upstream monitoring of the River Brue will be used to identify potential contamination events or changes in water quality prior to abstraction. Monitoring of the Glastonbury Reservoir will also be conducted to ensure the management of the reservoir maintains a stable water quality throughout the year.

To improve the water quality management, the Glastonbury Reservoir will incorporate an adaptive management system supported by real-time monitoring and predictive modelling. Monitoring buoys equipped with sensors can continuously measure key parameters such as dissolved oxygen, temperature and nutrient concentration (Envirotech, 2026).

Data will feed into operational systems controlling aeration, abstraction depths and ecological releases. For example, if rising water temperatures and increasing chlorophyll levels indicate the early development of an algal bloom, the system could automatically increase aeration or adjust abstraction depths to maintain optimal water quality from the draw-off tower. Similarly, predictive weather data could be used to prepare the reservoir for periods of warm conditions. This integrated monitoring and management approach allows the reservoir to respond dynamically to changing environmental conditions and maintain stable water quality over the long term.

7.3.4 Algae Removal

As a contingency during severe bloom events, mechanical algae-removal systems may be deployed. Floating robotic algae skimmers operate by collecting surface accumulation using conveyor belts or mesh filtration, which lift the algae from the water surface and store it for removal. Removing the biomass directly prevents the recycling of the nutrients back into the water during decomposition, which reduces the likelihood of further blooms from occurring. The collected algae could also be used as a nutrient-rich organic fertiliser, creating an opportunity for resource recovery within the

catchment. Although these systems would not be running continuously, they would give an additional layer of protection within the reservoir for its water quality management strategy.

Through the combination of upstream and in-reservoir monitoring, the proposed reservoir can maintain a proactive approach that identifies issues before they impact the drinking water supply and can minimise the demand on the water treatment plant, ensuring a reliable water supply.

7.4 Floating Solar Panels

Floating solar panels, also known as floatovoltaics, can be incorporated into the reservoir design to provide renewable energy, allowing the scheme to be partially self-sufficient, providing energy for the other physical processes used in the design, all while supporting water quality objectives. The floating solar panels consist of photovoltaics mounted on buoyant platforms anchored to the reservoir bed. These solar panels have an additional benefit that they can reduce water evaporation through the shading of the water surface. Additionally, the cooling effect of the water can boost the efficiency of the panels by up to 15% (Ukpanah, 2025). Also, the floating solar panels have unobstructed sun exposure, unlike on land, where many obstructions usually stop full daylight exposure, allowing the panels to be exposed to the sun at all times. To avoid reducing dissolved oxygen levels or inhibiting natural processes, surface coverage will be limited. Combined with floating wetlands, total coverage will remain within the recommended 18–50%, with bubble plumes installed beneath solar arrays to prevent localised stratification.

8.0 Water Draw-off Tower

8.1 Requirements

The design of the water draw-off tower will be governed by crack-width limitation, durability, resistance to restrained deformation, material selection and embodied carbon. Thermal effects, shrinkage and creep resulting in restrained deformation are the primary cause of cracking in water draw-off towers. Crack width limitation is essential to ensure watertightness and durability of the structure.

Following CIRIA C766, table 2.1 (CIRIA, 2020), limiting crack widths for serviceability in water retaining structures are between 0.05mm and 0.2mm. For the tightness class, it is deemed that leaking should be limited, but surface staining is acceptable, which puts the structure in tightness class 1. The recommended maximum crack width corresponding to the height and thickness is 0.05mm, where the reinforcement must adhere to the minimum area of reinforcement required by EN1992-1-1 (BSI, 2023).

Minimum cover depth has been defined by the BS 8500 guide, table 6 (TCC, 2024). For at least a 100-year working life, in primary exposure class XC3/4 and secondary exposure class XF1, C45/55 concrete requires a cover depth of 30mm plus allowance for deviations (+10mm), resulting in 40mm total cover. Exposure class XC3/4 has been chosen, where cyclic wetting and drying occurs due to changing water levels, and class XF1 has been chosen, where the water level is fluctuating and allowing possible freeze-thaw attacks.

8.2 Material Selection

In selecting material, the key priorities were durability and sustainability. Currently, 75% to 98% of cradle-to-gate carbon is due to cement. To reduce the embodied carbon, common substitutions can be included in the mix design, like fly ash and GGBS. While GGBS has advantageous qualities such as low permeability, it is a highly utilised resource and a review discussed by the IStructE indicated that global clinker production is 8x to 12x higher than global GGBS production (IStructE, 2023). The

use of GGBS in the water draw-off tower would likely cause a reduction in its use elsewhere, resulting in no overall reduction in embodied carbon. Similarly, fly ash has advantageous properties with low permeability, but the supply is limited in the UK.

To achieve both high durability and low embodied carbon, a developing technology and new emerging cement are being proposed. The design will consist of an LC3 cement mix and innovative bacterial self-healing.

LC3's use of calcined clay and limestone additions in pozzolanic reactions results in a dense microstructure with low permeability. Due to its reduced clinker content, the total heat release during hydration is lower than that of Ordinary Portland Cement, reducing the risk of early age thermal cracking. LC3 makes use of abundantly available materials (clay and limestone) that can be sourced locally. It has been shown that LC3 performs just as well or even better than OPC and high strength properties of OPC can be achieved (Hosen and Chen, 2025). It is feasible to implement the use of LC3 as no specialist equipment is required and the lower clinker content reduces the cost compared to OPC. In terms of standardisation, combinations of calcined clay and limestone are accepted in EN197-1 down to 65% clinker content. There are many existing applications of LC3, and Glastonbury Reservoir can be the next in tackling the adoption of low-carbon concrete design.

Since LC3's reduced clinker content limits its self-healing ability for microcracks, less than 0.3mm, bacterial self-healing will be utilised for healing small cracks, improving the durability of the structure and reducing the need for manual crack maintenance. The application in a reservoir is ideal as water entering cracks will activate the bacteria, which produce calcium carbonate through microbial induced calcite precipitation (MICP). Studies have shown that MICP technology is a viable approach in enhancing the durability performance of LC3, demonstrating a self-healing rate of up to 97% in optimal conditions as well as an enhanced compressive strength after 28 days (Wei et al., 2024). Its initial cost is justified by reduced maintenance costs and improved durability, ensuring the structure has a design life of over 100 years. In addition, the self-healing properties justify a relaxation on reinforcement demand for crack control.

8.3 Tower Design

The reinforced concrete section of the tower, as shown in Figure 12, will be of circular geometry with a height of 15.5m of which 14.0 m are designed for tightness class 1. The top of the tower for access and the control room will be constructed out of timber and is 2.2m high for comfortable headroom, resulting in a total tower height of 17.7m. The towers internal diameter of 5m with a 300mm wall thickness will allow for comfortable access and working space when required. It will house a suction pipeline of outer diameter of 630mm with a wall thickness of 57.27mm leading to the pumping station, which will discharge the water into the pipeline system to transfer the water to Honeyhurst WTW.

The LC3 mix design will achieve strength class C45/55, consisting of 380kg/m³ cement content, a w/c ratio of 0.40 and limestone aggregate. SikaGrind CC admixture is recommended to mitigate the water demand due to calcined clays and improve the workability (Sika, 2026). Limestone aggregate has been selected due to its numerous advantages, including good strength, low risk of alkali-silica reaction and decrease in drying shrinkage (Aquino et al., 2010).

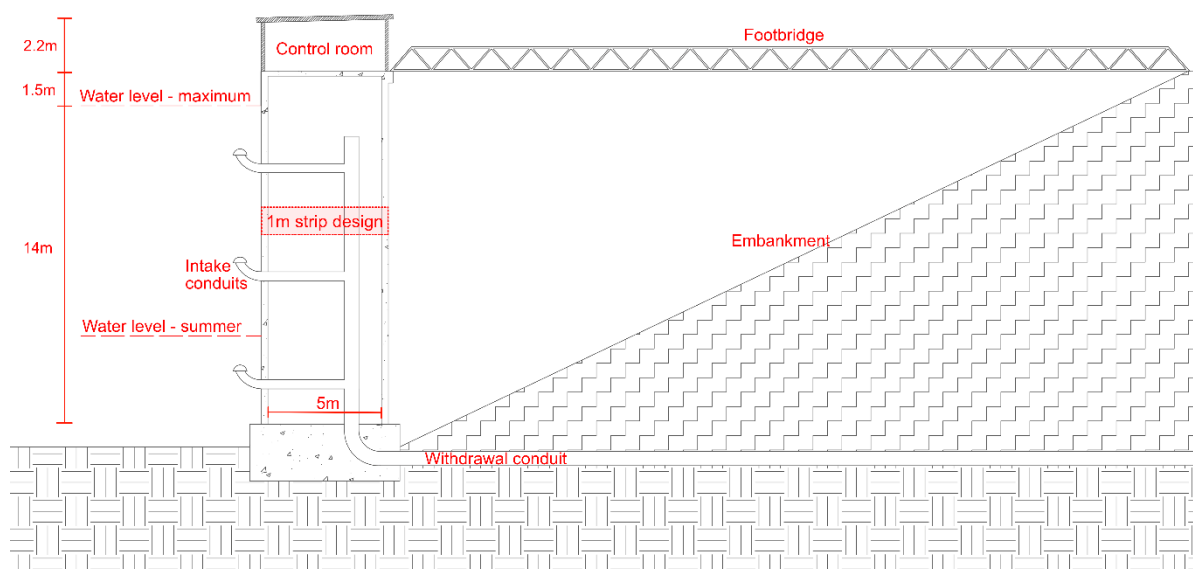


Figure 12 - Water tower schematic diagram

Upon designing the reinforcement, it was determined that achieving a crack width of 0.05mm with reinforcement alone was impractical and required a high amount of reinforcement, which would increase embodied carbon. Therefore, the design will utilise a multi-method crack control strategy. The reinforcement checks aim to limit crack widths to 0.2mm with additional strategies as follows: the LC3 mix will generate less heat than OPC; the limestone aggregate has a low coefficient of thermal expansion, reducing drying shrinkage; bacterial self-healing concrete will fill cracks up to 0.3mm, increasing durability and allowing some lenience on reinforcement demand; concrete placing to take place during cooler seasons; layered pouring to reduce internal temperatures and steel formwork to permit rapid cooling. Mitigations during placing are to be further discussed with the relevant contractor to develop a full early-age thermal crack control plan.

The maximum temperature drop, T_1 of the designed concrete mix, which has a great effect on thermal cracking, was determined using CIRIA C766 spreadsheets. Inputs and outputs can be seen in Table 6.

Table 6 - Inputs and output for CIRIA C766 temperature spreadsheet

Inputs	Values	Units	Comments
Binder Content	380	kg/m ³	
Binder Type	GGBS		LC3 similar to 50% fly ash substitution but GGBS has been selected for a conservative approach.
Addition	50	%	
Density	2400	kg/m ³	
Specific Heat	1	J/kg °C	
Pour Thickness	300	mm	Lower thickness results in lower T1
Formwork Type	Steel		Permits rapid heat loss, reducing T1
Wind Speed	4	m/s	Typical UK conditions
Formwork Removal	36	hours	LC3 has lower early strength
Thermal Conductivity	1.8	W/m °C	
Placing Temperature	20	°C	
Mean Ambient Temperature	15	°C	
Output			
Temperature Drop (T1)	14	°C	

The reinforcement required to achieve a maximum crack width of 0.2mm following the calculated temperature drop and assuming full restraint of 1.0 due to the cylindrical shape was calculated using the CIRIA C766 spreadsheets. Inputs and outputs can be seen in Table 7. The design was heavily governed by end restraint.

Table 7 - Inputs and outputs for CIRIA C766 restraint spreadsheet

Inputs	Symbol	Values	Units	Comments
Section thickness	h	300	mm	
Strength Class	$f_{ck}/f_{ck,cube}$	C45/55		Higher strength allows lower cover, beneficial for crack control
Age at cracking	t_c , short term	3	days	
	t_c , long term	28	days	
Creep factor	K_1	0.65		
Sustained load factor	K_2	0.8		
Coefficient of thermal expansion of concrete	α_c	9.5	$\mu\epsilon/^\circ\text{C}$	For limestone aggregate
Characteristic yield strength of reinforcement	f_{yk}	500	Mpa	
Elastic modulus of reinforcement	E_s	200	GPa	
Temperature drop	T_1	14	$^\circ\text{C}$	
Restraint	R	1		
Coefficient for bond characteristics	k_1	1.14		
Cover	c	40	mm	30mm required cover plus 10mm for deviations
Bar diameter	ϕ	14	mm	
Bar spacing	S	50	mm	Spacing is halved to account for two layers of reinforcement @ 100mm spacing.
Output				
Early age crack width	$w_{k,early-age}$	0.12	mm	End restraint governs
Long term crack width	$w_{k,long term}$	0.2	mm	End restraint governs

With two layers of reinforcement consisting of 14mm diameter bars and 100mm centre to centre spacings, the maximum crack width is limited to 0.2mm which bacterial self-healing concrete can comfortably heal. The reinforcement arrangement is displayed in Figure 13. Both layers of reinforcement lie within the surface zone, h_{ef} , defining the effective area of concrete in tension.

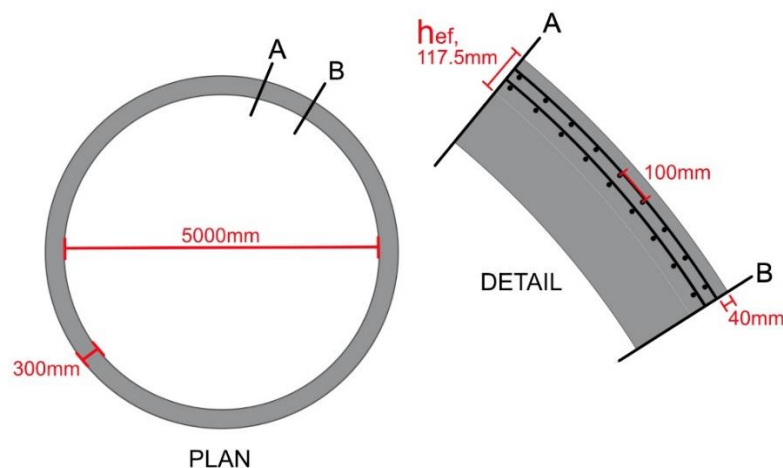


Figure 13 - Draw-off tower reinforcement arrangement detail

8.4 Embodied Carbon Optimisation

The embodied carbon optimisation of the structure was heavily driven by material choices, choosing materials low in carbon as well as materials that reduce the value of T_1 which in turn reduced reinforcement demand. The use of LC3 as a low carbon cement with the addition of bacterial self-healing reduces the high reinforcement demand for crack control and increases durability for over a 100-year design life, as well as utilising local material supplies of calcined clay and limestone. Figure 14 shows a comparison between the selected LC3 concrete and a CEM I mix with identical reinforcement, where embodied carbon calculations for stages A1-3 were carried out. The LC3 mix achieves $232\text{kgCO}_2/\text{m}^3$, whereas the CEM I mix achieves $387\text{kgCO}_2/\text{m}^3$. In reality, the CEM I mix would require greater reinforcement to control cracking due to higher heat release, making it an even poorer choice. The LC3 mix and required reinforcement fall comfortably in the 'A' class for embodied carbon according to the ICE Low Carbon Concrete Routemap (ICE, 2023. Low Carbon Concrete Routemap. London: Institution of Civil Engineers.).

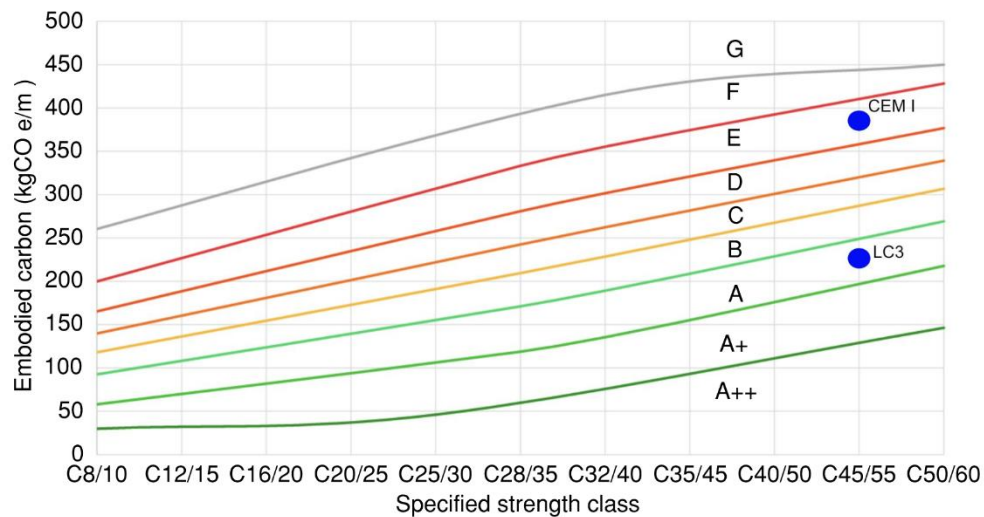


Figure 14 - Benchmark ratings for embodied carbon (ICE, 2023)

Optimisation was also considered when selecting the concrete strength and wall thickness. The thinner wall thickness with higher strength (C45/55) required less cover depth and less reinforcement. In contrast, the lower strength concrete (C35/45) with a thicker wall required greater cover depth and greater reinforcement. The differences between the Portland cement required in each mix and the differences in reinforcement requirements resulted in approximately the same embodied carbon. The design proceeds with the higher strength mix due to its greater durability from the lower w/c ratio. This confirms that the proposed material system meets both structural and sustainability objectives while achieving compliance with durability requirements.

9.0 Distribution-line System

9.1 Pipeline System

The Glastonbury Reservoir is located 13.4km away from the proposed site of the Honeyhurst WTW, where a 16.5km pipeline will transport the water. The system is designed to supply a peak flow of 20 ML/day in June-July and 5 ML/day for the rest of the year, where varying reduced flows will not have large impacts on efficiency. Pipeline design focused on minimising head losses through route selection and pipe details.

9.1.1 Route

Figure 15 shows the pipeline route selected for initial design. The route follows a flat contour, minimising elevation changes to only necessary areas, while minimising bends. The route also avoids passing through major towns and villages, as well as sites of important ecological value. The route is from the draw-off tower, where the water will flow 500m to the pumping station and then be pumped 16km through a rising main to the Honeyhurst WTW. Due to the location, major crossings are unavoidable. The route crosses two major roads and both the River Brue and the River Axe.



Figure 15 - Pipeline route

9.1.2 Pipeline Construction

The design of the rising main has been developed with reference to the Water Supply (Water Fittings) Regulations 1999 (HMSO, 1999) and the Pipelines Safety Regulations 1996 (HSE, 2012), ensuring compliance with relevant UK regulations. Health and safety considerations, including trench stability, safe access, and risks from existing buried services, will be addressed through the Construction (Design and Management) Regulations during detailed design.

The pipeline will be buried at an average depth of approximately 1 m, with a tolerance of ± 0.25 m. This provides adequate protection from vandalism and reduces the likelihood of accidental third-party damage. It will also prevent freezing of water inside the pipes and help maintain a consistent hydraulic profile by avoiding abrupt changes in pipe geometry. A minimum cover of 0.75 m will be maintained in all locations, increasing to 1.2 m where the pipeline crosses roads or trafficked areas, as required under the Water Supply (Water Fittings) Regulation 1999 (HMSO, 1999).

For the majority of the route, installation will be undertaken using traditional cut-and-cover trenching, with pipe sections joined and placed into the trench. The trench will include appropriate bedding and surround materials in accordance with water-industry practice, typically comprising sand or fines to prevent point loading on the pipe. Backfilling and reinstatement will follow local authority requirements.

Localised constraints along the route will require additional construction methods. Short sections may require diversions, night-time working, or traffic management to minimise disruption when crossing existing roads or infrastructure. Where the pipeline crosses watercourses or where open-cut methods would be impractical, horizontal directional drilling will be used to install the pipe with minimal environmental impact.

9.1.3 Pipe Details

High-Density Polyethylene (HDPE) pipes have been selected for the rising main due to their wide availability in the UK market, ensuring reliable supply chains and cost-effective procurement. It is a proven material for water distribution systems, offering strong resistance to both internal and external corrosion, as well as long-term durability when buried, with a typical service life of over 100 years. The flexibility of HDPE makes it particularly suitable for ground movement and settlement, while burial protects the pipe from UV degradation.

For this scheme, an HDPE pipe with a Standard Dimension Ratio (SDR) of 11 has been assessed as the preferred option, suitable for high-pressure distribution applications. SDR 11 corresponds to a PN16 pressure rating under BS EN 12201-2:2024, Table A-1 (BSI, 2024), which defines pipe pressure class based on long-term creep strength. The nominal pressure (PN) classification represents the maximum allowable continuous operating pressure for potable water systems, with an applied factor of safety of 1.25. The SDR value represents the ratio between the pipe's outside diameter and wall thickness, and therefore directly determines its long-term pressure capacity.

The discharge pipe has a length of roughly 16km from the pumping station to the exit at Honeyhurst WTW, with an outer diameter of 500mm and a wall thickness of 45.45mm. This results in internal flow velocities of approximately 1.76 m/s at the peak flow rate of 20 ML/day and 0.44 m/s at the typical flow rate of 5 ML/day, for a continuous 24-hour flow. These velocities fall within the recommended range of 0.3–2 m/s, ensuring that sediment deposition and biofilm formation at low velocities are prevented and stops cavitation and high friction losses at high velocities.

9.2 Pipeline Model

Hydraulic calculations were carried out to determine the Total Dynamic Head (TDH) for the proposed 16 km rising main from Glastonbury Reservoir to Honeyhurst WTW. The purpose of this analysis is to establish the required pump characteristics and to confirm that the proposed pipeline detail is suitable for the expected operating pressures. A summary of the head losses for both the peak and typical flow conditions is presented in Table 8.

Table 8 - Pipeline head losses

Required Flow		20 ML/day	5 ML/day
Static Head	h_s	7 m	7 m
Total Minor Losses	h_m	0.42 m	0.03 m
Total Friction Losses	h_f	80.30 m	6.52 m
Total Dynamic Head	TDH	87.72 m	13.54 m

The static head of 7 m results from the elevation difference between the reservoir (3 mAOD) and the WTW (10 mAOD). Minor losses are small in comparison with friction losses, as the pipeline is assumed to follow a largely straight alignment with minimal sudden geometry changes. It is assumed that water enters Honeyhurst WTW at atmospheric pressure, as the plans for the WTW are not detailed as of this proposal.

Friction losses were calculated using the Darcy-Weisbach equation with a friction factor determined from the Colebrook-White formula. These losses dominate the TDH due to the long length of the rising main. For the peak flow of 20 ML/day, the total dynamic head is 87.7 m, reducing to 13.5 m for the typical flow of 5 ML/day.

This pipeline model is intended for preliminary design only. A more detailed hydraulic model will be required to refine the route, confirm the number and location of fittings, and assess unique hydraulic behaviours along the 16 km alignment. As the highest TDH only occurs for approximately two months of the year, the pumping system will be configured to meet the peak demand while also operating efficiently during typical flow conditions.

9.3 Pump Selection

A single pumping station at the reservoir is proposed to supply the water with the required head to reach the WTW. For this preliminary sizing, a pump catalogue supplied by Grundfos, as recommended by the client, was used. For the peak flow requirements, it was found that a parallel

system would be necessary, as the required flow rates were too great for a single pump, however, this allows for a more efficient operation of a single pump during typical flows.

9.3.1 Pump Characteristics

For the maximum demand of 20 ML/day, corresponding to a peak duty head of 87.7 m for a flow of 833m³/h for a continuous 24hr period, a configuration of four Grundfos CRN 215-3 pumps (Grundfos, 2026) operating in parallel was identified as suitable, where one of these units is always on standby for allowable redundancy. A combined characteristic curve for this configuration is presented in Figure 16, showing a working pump efficiency of 77.6%. The selected pump model is a vertical multistage inline centrifugal pump, working at a maximum PN16, so as not to damage the pipes. Inline suction and discharge allow for integration without significant changes to the pipe alignment. The multistage configuration provides stable head generation across a wide operating range, making it suited for variable seasonal demand.

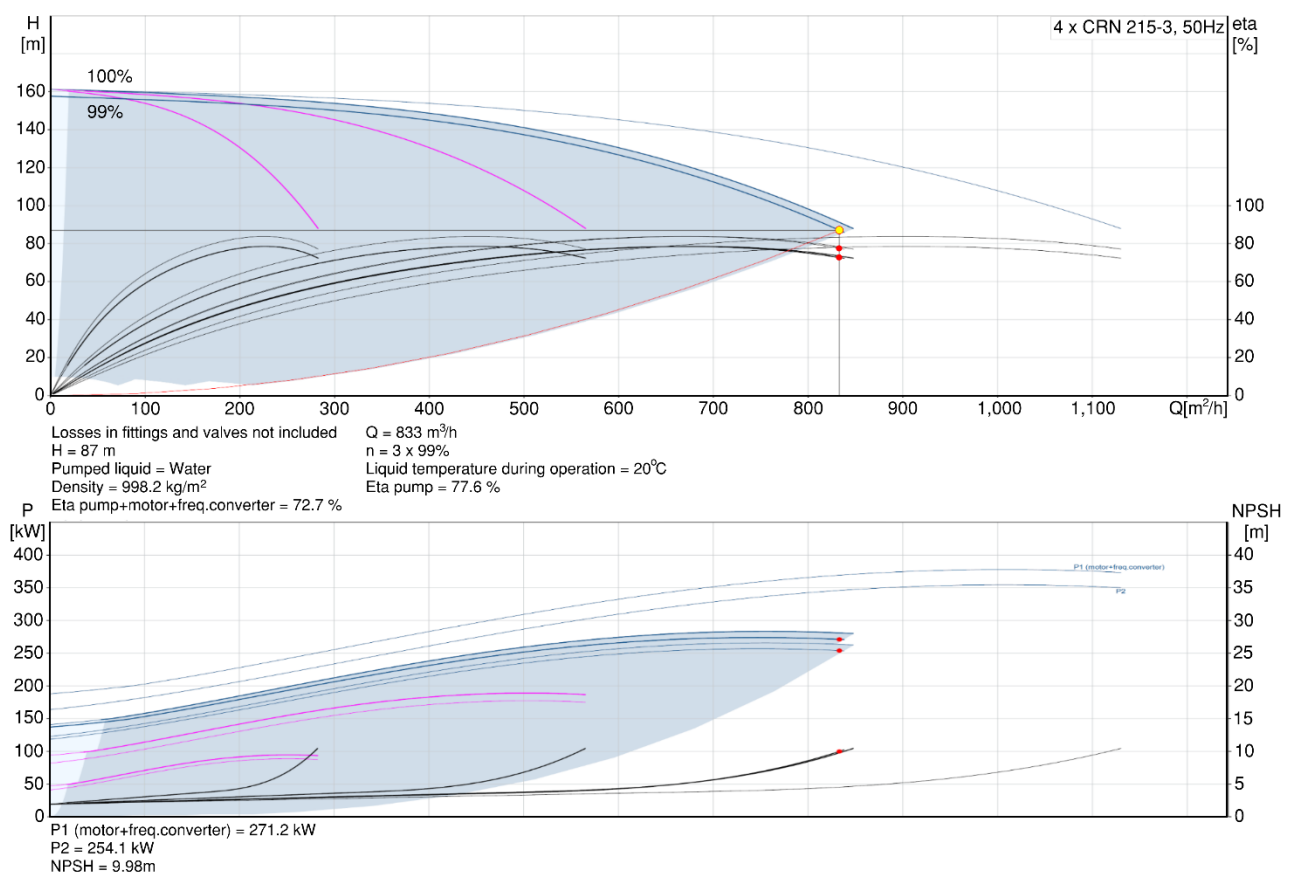


Figure 16 - 4xCRN 215-3 pump characteristic chart (Grundfos, 2026)

For typical operation at 5 ML/day and a duty of 13.5 m, analysis showed that a single CRN 215-3 pump operating under variable-speed control would be capable of achieving the required flow rate. Efficiency under these reduced-flow conditions is reduced, but preferred over an additional separate low-flow pump, as it avoids the operational complexity associated with switching between pump sets, reduces spare-parts requirements, and simplifies maintenance planning.

This configuration meets both peak flows and typical flow conditions, whilst offering strong operational resilience through standby capacity, and maintains efficient performance across variable flow rates depending on required demand. However, there is potential to optimise the system further. Research in detailed design should assess the use of custom-built pumps to achieve the variable flows at an optimum efficiency.

For this scheme, continuous 24-hour pumping was selected as the preferred operational mode, as it provides a more efficient and manageable solution compared with a shorter daily operating window. Operating continuously allows the pumps to be sized significantly smaller, reduces cost and reduces friction losses in the rising main due to lower average flow velocities. It should also be noted that this design is to manage the maximum output of 20 ML/day, where lower flows are expected until demand grows over time.

9.3.2 Pump Cavitation Checks

A cavitation check was undertaken for the proposed pump by comparing the manufacturer's required Net Positive Suction Head (NPSH_r) with the calculated Net Positive Suction Head available (NPSH_a) at the pump suction. The assessment shows that the pump requires an NPSH of 10.21 m at the duty point, while the system provides an NPSH_a of 10.33 m. This demonstrates that the pressure at the pump inlet remains above the vapour pressure of the fluid under operating conditions, thereby avoiding cavitation and ensuring stable, reliable hydraulic performance.

9.3.3 Pipeline Pressure Checks

Pressures were checked along the pipeline using the pipeline long section, shown in Figure 17. At pump discharge, the resulting maximum pressure head is 89m supplied by the pumps, which is within the PN16 pipework rating. Along the rising main, the highest elevation occurs at around 1.7 km, where the pressure has the greatest reduction, however, the minimum internal pressure head remains at 73m, comfortably above vapour pressure and therefore not at risk of cavitation. Pressures at downstream remain safe, with no major changes in elevation, ensuring the entire system operates safely within PN16 limits while maintaining adequate positive pressure throughout the pipeline.

It is assumed that the pumps will use variable speed to reduce the effects of hydraulic transients, mitigating surges within the pipeline. A surge assessment will be required at detailed design to ensure that transient pressures remain within safe working conditions during failure.

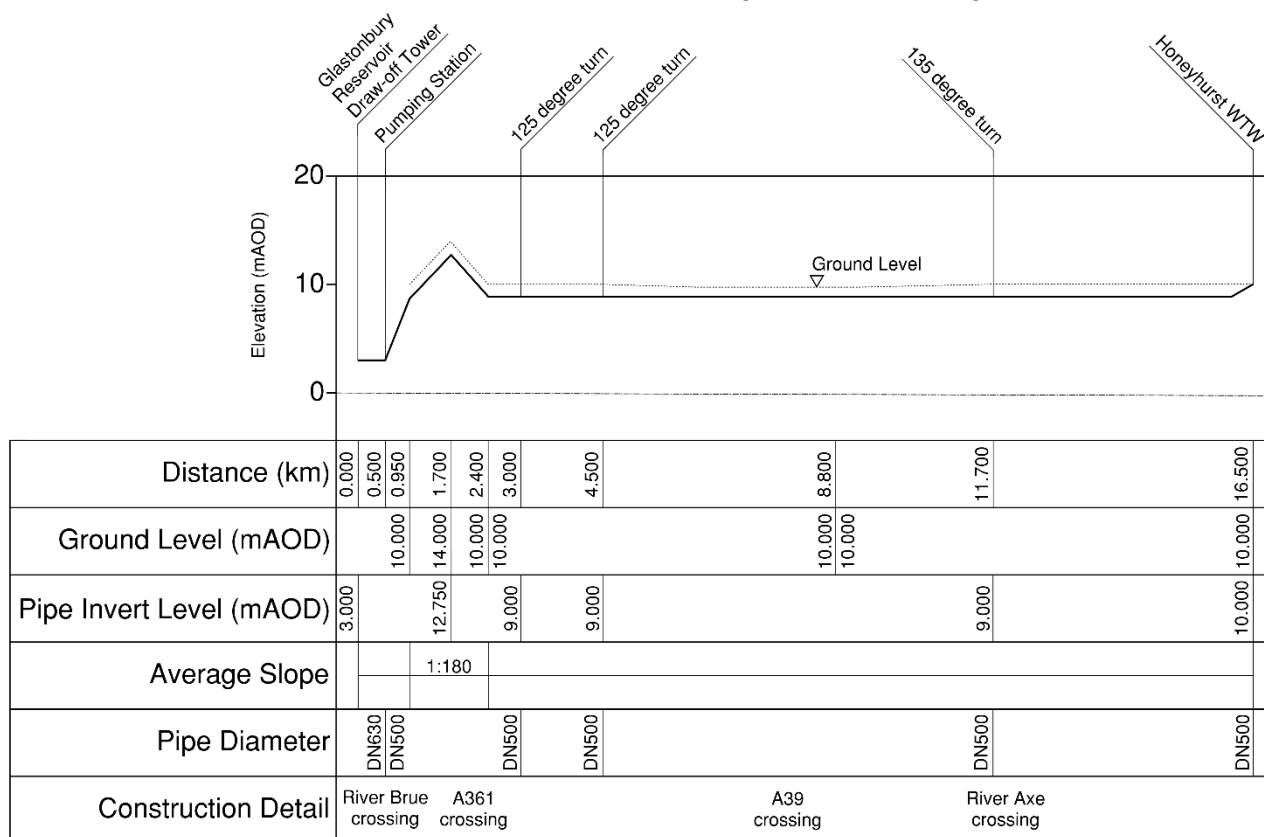


Figure 17 - Pipeline long section

9.4 Cost Assessment

An approximate cost assessment was undertaken covering construction and the first 40 years of operation. The analysis compared the proposed scheme against the minimum scoping option to determine the cost implications of delivering a higher-capacity, future-resilient system. While the proposed design represents a 13.5% increase in total cost, it delivers approximately 2.5 times the output, enabling long-term operational flexibility and improved resilience. The cost breakdown, presented in Figure 18, indicates an estimated £5.7 million whole-life cost for the proposed option under current energy rates (18.5p/kWh), with sensitivity testing showing higher costs of £6.4 million where elevated electricity prices are assumed (25p/kWh). This comparison demonstrates that the significant increase in output justifies the additional investment, as the leading cost is the pipeline installation, which is the same for both scopes.

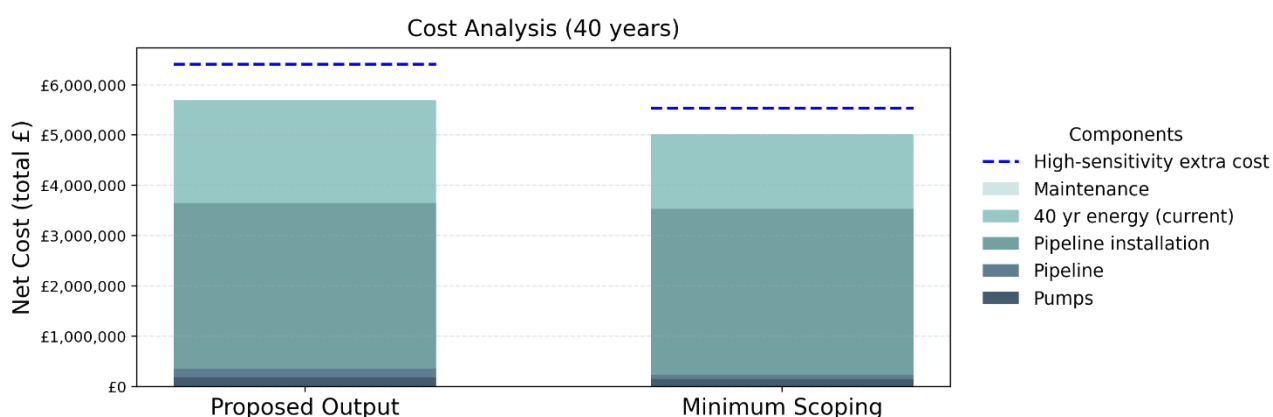


Figure 18 - Pipeline system cost assessment (40 years approx.)

10.0 Conclusion

Glastonbury Reservoir provides a resilient and sustainable solution to long-term water-resource challenges in the Bristol Water supply region. Hydrological modelling based on a 1000-year flow record demonstrated that the River Brue can reliably support abstraction well beyond the minimum project requirement. This surplus enables the reservoir to meet the scoped output of 8 ML/d in June-July and 2 ML/d during the rest of the year, while also providing enhanced capacity of up to 20 ML/d and 5 ML/d respectively, supplying approximately 14,000 households and supporting future population growth.

The final reservoir design balances water supply with environmental considerations. A storage volume of 3 million m³, combined with controlled operating rules, allows the reservoir to deliver flood-alleviation benefits by increasing abstraction during high-flow events, reducing downstream hydraulic stress, and lowering the likelihood of road and farmland flooding. Conversely, during low-flow periods, controlled releases of approximately 2 ML/d support the ecological health of the Westhay Moor wetlands, demonstrating the reservoir's role as a multifunctional catchment-management asset.

Water quality is protected through a combination of design measures and adaptive management strategies. These measures minimise algal growth, reduce treatment demand at Honeyhurst WTW, and improve ecological resilience. The 16 km rising main, efficient pumping system, compliant embankment and draw-off tower structure ensure long-term operational reliability. Overall, the scheme provides a technically robust, environmentally responsible and future-proof contribution to regional water security.

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